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MODULE III

Principle of Synchronous Motor, Equivalent circuit, Phasor diagrams, Power flow diagram and equations, Losses and efficiency - Numerical Problems, Power-angle Characteristics, V Curve and Inverted V Curves.

Three phase Induction motor – Constructional features, Expressions for Power and Torque-Torque-Slip characteristics, Phasor diagram, Equivalent Circuit of Induction motor- Tests on Induction motors for determination of equivalent circuit-Numerical Problems.

Synchronous motor

A synchronous motor is an electrical machine which converts AC electrical energy into mechanical energy

An Alternator can work as a Synchronous motor by connecting a 3 phase AC supply to stator (armature)

A synchronous motor runs at synchronous speed, $N_s = (120f)/P$

Speed of rotation depends on stator supply frequency (f) & number of poles in stator (P). For a machine, number of poles is a constant. So $N_s \propto f$

Construction of Synchronous motor

A synchronous motor is an electrical machine that runs at synchronous speed & converts electrical energy into mechanical energy

It is fundamentally an Alternator operated as a motor
It has mainly 2 parts

1. Stator
2. Rotor

1. Stator or Armature

Stator is the stationary part

Main parts of **stator** are **stator core**, **stator winding** & **stator frame**

Stator core contain an iron ring made of silicon steel laminations

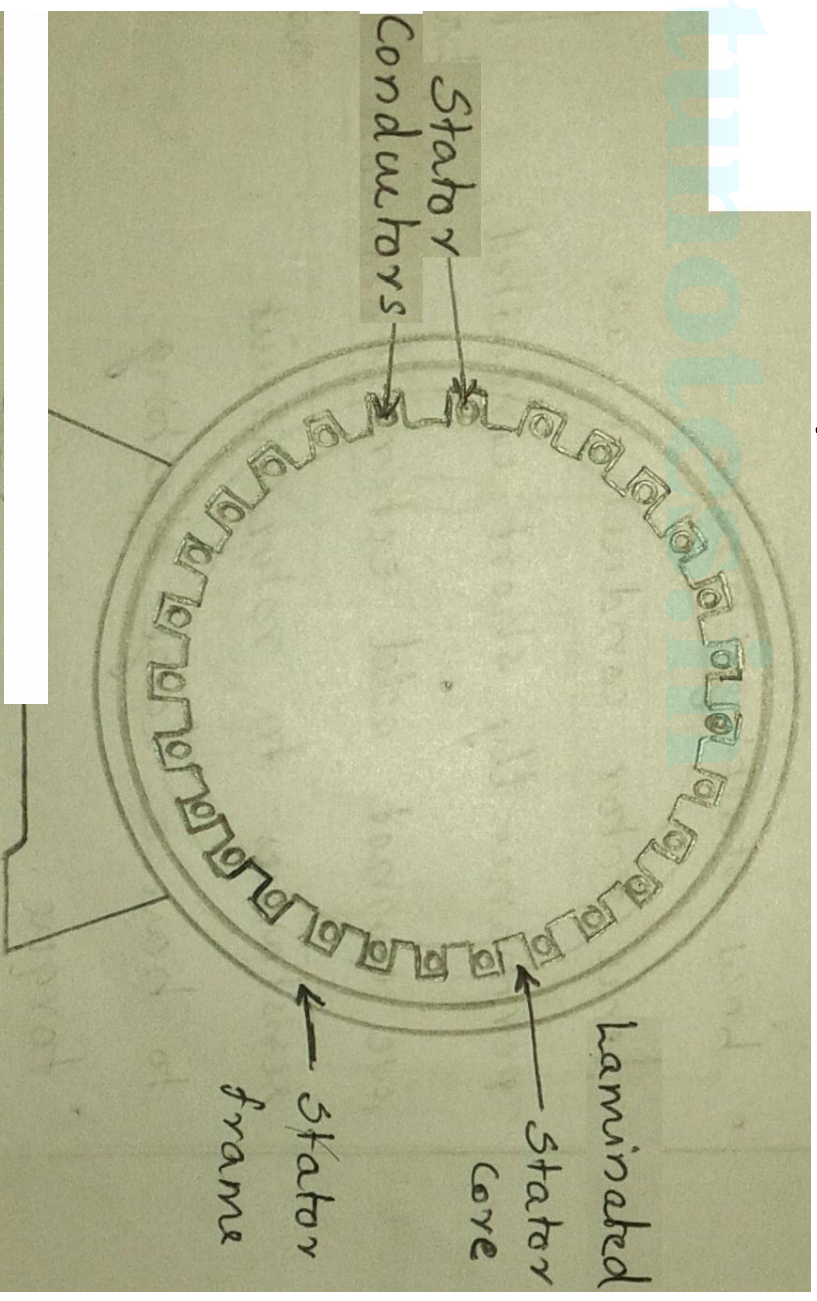
Slots are cut on the inner periphery of the stator core, in which stator conductors are placed

Stator core is laminated to reduce eddy current loss

It also contain axial & radial ventilating ducts to provide efficient cooling

Slots can be open, semi closed or totally closed

Open slots are commonly used



Stator contain **a 3 phase winding** which is normally star connected
Stator conductors are made of copper or aluminium
Windings can be single layer or double layer, lap or wave, concentrated or distributed

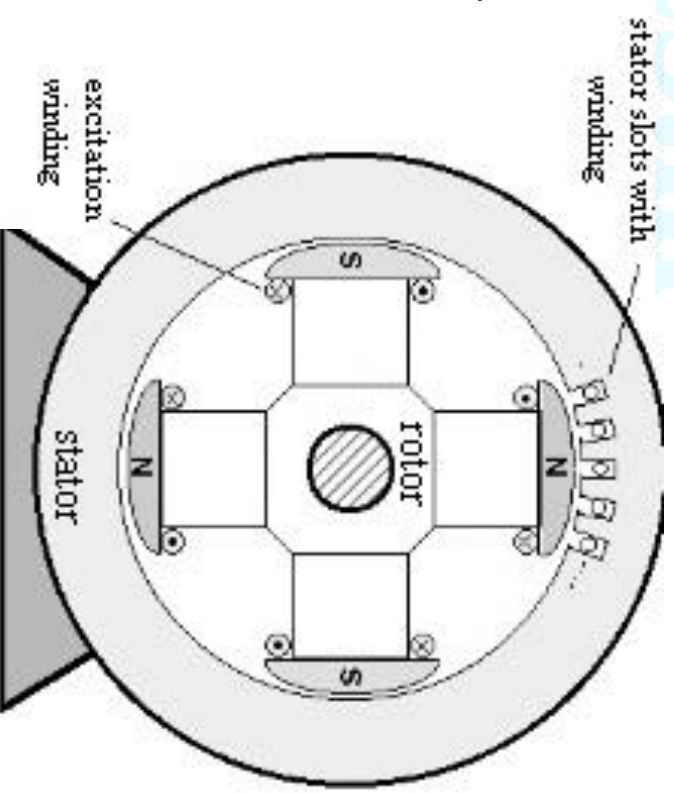
The whole structure is held in a **frame** made of cast iron or welded steel plate

b) Rotor or Field magnet system

It is the rotating part of the machine
Rotor contain Field magnets & are supplied with DC voltage

For this external DC sources are used & exciting current is supplied to rotor through slip rings & brushes

An exciter mounted on the motor shaft can also be used to provide necessary excitation.



- Normally, the rotor is having a salient pole structure
- Field poles contains laminated pole core & pole shoe (to reduce eddy current losses) and field windings

Salient features of Synchronous motor

- It runs at Synchronous speed or it won't rotate. The only way to change the speed of motor is to change the supply frequency
- Synchronous motors are generally of salient pole type
- It can be made to operate at different power factors by varying the field excitation
- Synchronous motor is not self starting. Auxiliary means has to be used for starting the motor.

Principle of operation

- When a 3 phase supply is given to the stator, a rotating magnetic field is produced which rotates at synchronous speed
- When a DC supply is given to field windings, stationary N & S poles are created in rotor

- Now we got a revolving magnetic field poles in stator & stationary poles in rotor

Suppose at any instant, let position of stator & rotor magnetic poles be as shown in figure

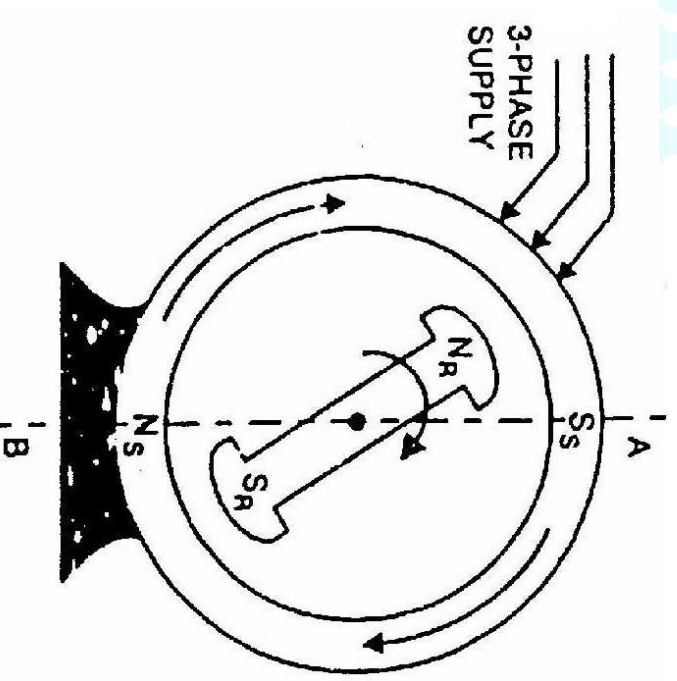
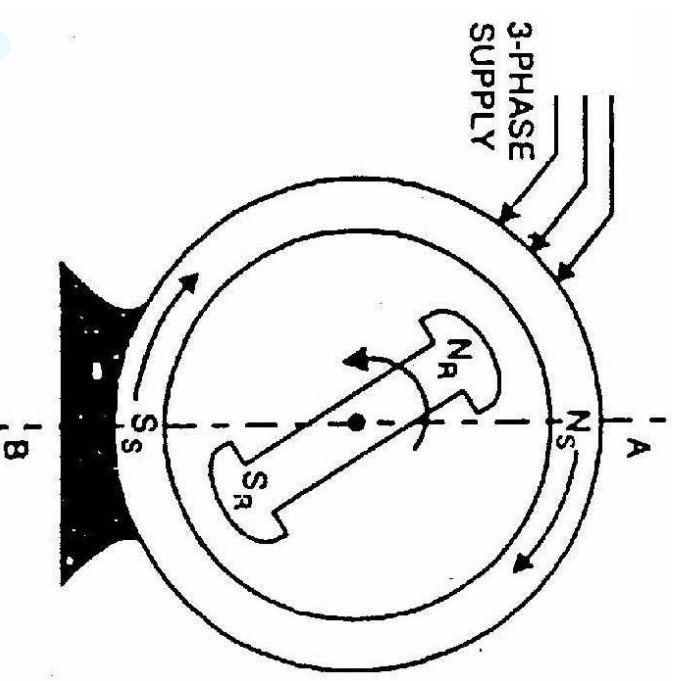
Now stator North pole (N_s) & rotor North pole (N_r) will repel each other & rotor experiences a torque in anticlockwise direction

After a period of half cycle, stator poles changes their position but rotor poles remain at previous position

Now stator South pole (S_s) & rotor North pole (N_r) will attract each other & rotor experiences a torque in clockwise direction

Since stator poles changes their polarity rapidly, rotor experiences torque in opposite directions in a cycle.

Due to high inertia of rotor, motor fails to start



How to get unidirectional torque?

If rotor is rotated by some external means at a speed near synchronous speed, in the same direction of stator magnetic field, then rotor will experience a unidirectional torque

Assume that rotating magnetic field in stator is rotating in clockwise direction & rotor is also rotated in same direction by some external means.

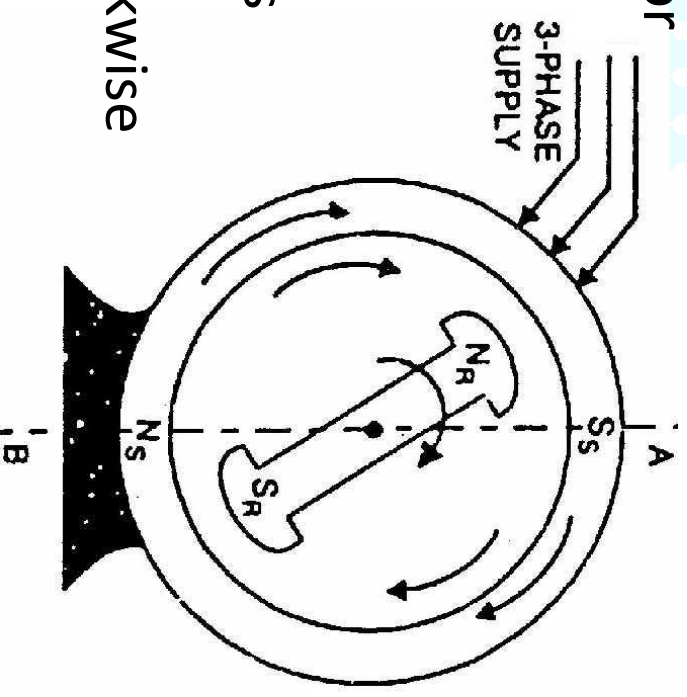
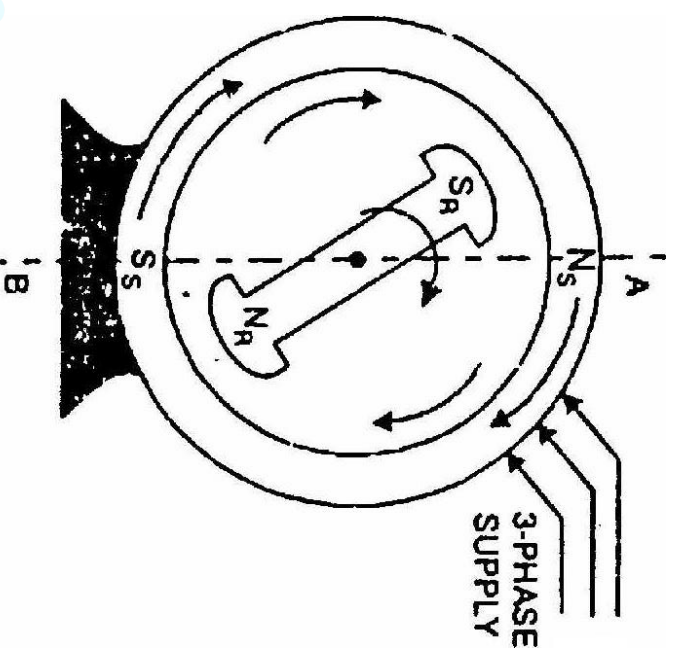
Now both stator & rotor magnetic poles are rotating in same direction at almost same speed

At any instant let the position of stator & rotor magnetic filed poles be as shown in figure

Now stator South pole (S_s) & rotor North pole (N_r) will attract each other & rotor experiences a torque in clockwise direction

After a period of half cycle, stator & rotor magnetic field poles changes their position as shown in second figure

Now again rotor experiences a torque in clockwise direction.



- Hence a continuous unidirectional torque is produced
- Under this condition, stator & rotor poles of opposite polarity experiences a strong attraction & the mutual attraction locks the rotor & stator together

- Now if the external prime mover is removed, rotor will continue to rotate at synchronous speed due to magnetic locking between stator & rotor

Equivalent circuit of a Synchronous motor

- In a synchronous motor we are giving two supply, DC supply to rotor & 3 phase AC supply to stator

- When AC supply is given to stator, a current flows through stator windings

- Stator windings have resistance (R_a) & synchronous reactance (X_s)

- Also when a DC supply is given to rotor, a magnetic field is produced & it links stator conductors,

- As rotor rotates, flux linked with stator conductors changes & an EMF induces in stator windings. This is called back EMF (E_b) or counter EMF

- The back EMF opposes the stator supply voltage

Back EMF depends on speed of rotor & flux in rotor

Since speed of rotor is constant (synchronous speed), $E_b \propto \text{flux}$ & flux depends on field current (I_f)

Stator supply voltage produces a rotating magnetic field & it links rotor. Since rotor is also rotating at synchronous speed, no EMF induces in rotor

Now the equivalent circuit of Synchronous motor is given as

From equivalent circuit, net voltage per phase in stator winding is given

$$\text{as } E_r = V - E_b$$

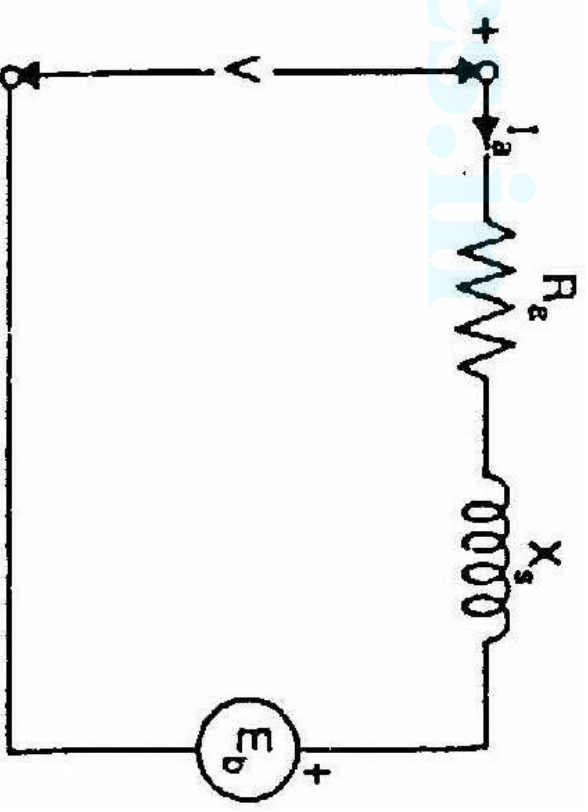
Armature current/phase

$$I_a = E_r / Z_s$$

Note

Synchronous motor is said to be

normally excited if $V = E_b$. If $E_b < V$, motor is said to be under excited & if $E_b > V$, motor is said to be over excited



Starting methods of Synchronous motor

Synchronous motor has no self starting torque & motor must first be brought upto synchronous speed by some external means for getting unidirectional torque in rotor.

Various methods of starting of synchronous motors are

1. From DC source

If a DC supply & a DC motor (shunt/compound) is available, the

Synchronous motor is coupled to DC motor & is started using DC motor

First the DC motor is coupled to Synchronous motor

The DC motor is started & is rotated at synchronous speed

AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)

Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

The DC motor is now disconnected from supply

2. By using an AC motor

Here a small AC induction motor (called pony motor) is used for starting the Synchronous motor

First the induction motor is started & will rotate Synchronous motor rotor at near synchronous speed

AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)

Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

The Induction motor is now disconnected from supply

3. By providing damper windings

Synchronous motor can be made self starting by providing a special winding on rotor poles known as damper winding or squirrel cage winding

Damper winding consist of short circuited copper bars placed in rotor poles

When AC supply is given to stator, the synchronous motor will start as an Induction motor & rotates at a speed near synchronous speed

Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

Motor on load

In a DC motor or Induction motor, as load increases, speed decreases

Decrease in speed reduces back EMF &

additional current ($I_a = (V - E_b)/R_a$) can be drawn from supply to meet increase in load

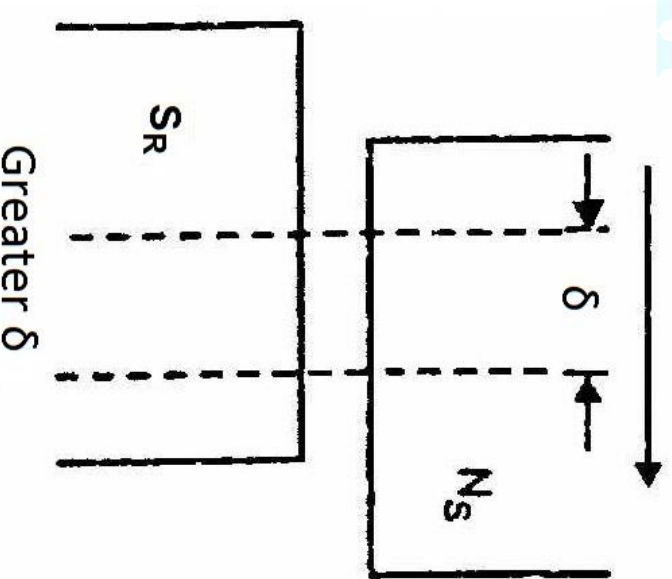
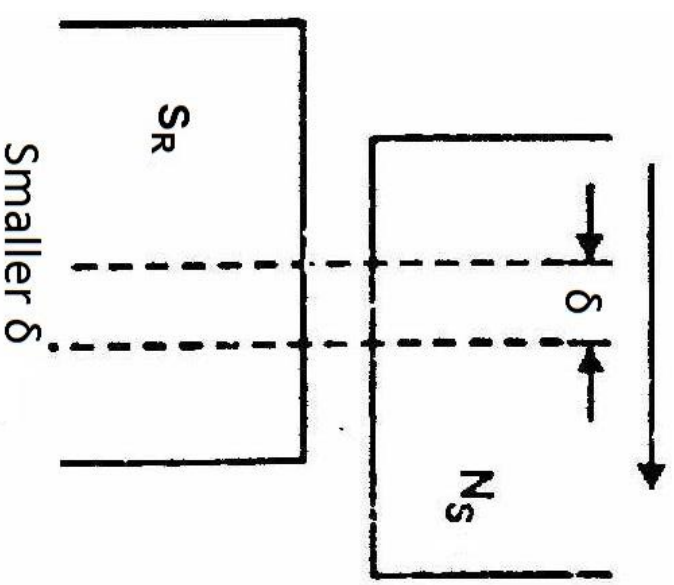
But a Synchronous motor runs only at synchronous speed

When load increases, the rotor falls back the stator poles by some degree & continue to rotate at synchronous speed

i.e, as load increases, load angle(δ) increases

When load angle changes, phase of back EMF changes w.r.to supply voltage(V)

As a result the net voltage (E_r) increases & consequently armature current (I_a) increases



Pull out Torque

As load on Synchronous motor increases, load angle(δ) increases, but motor continue to rotate at synchronous speed

If δ increases above a certain value, rotor gets pulled out of synchronism & motor comes to stand still

The load torque at which the motor pulls out of synchronism is called ***pull out torque or breakdown torque***

Phasor diagram of a Synchronous motor

Consider a Synchronous motor. Let, V = stator supply voltage/phase

E_b = Back EMF/phase

Z_s = Synchronous impedance/phase

Motor on no load –

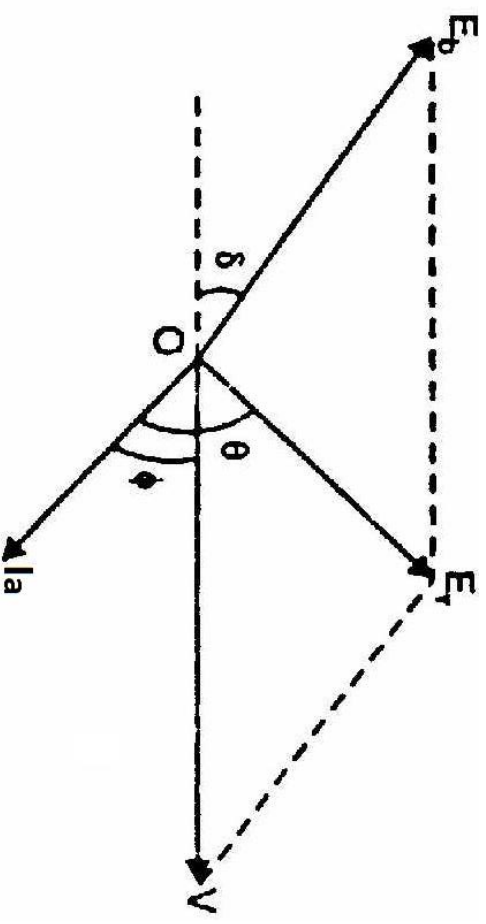
On no load, load angle(δ) is small

E_b falls back V by a small angle δ

Net voltage/phase = E_r

Armature current, $I_a = E_r / Z_s$

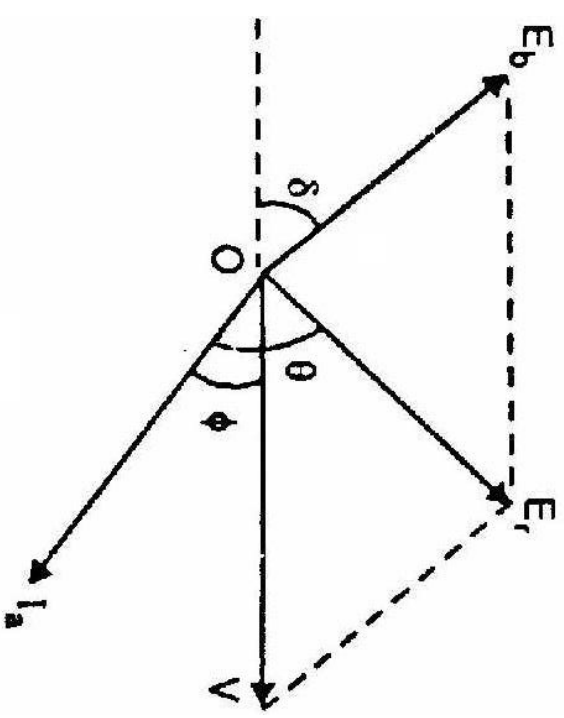
I_a lags behind E_r by an angle $\approx 90^\circ$ (Since R_s is very small, $Z_s \approx X_s$)



Motor on load –

When load increases, δ increases

Power input/phase, $P_i = VI_a \cos\phi$



Effect of varying excitation on

Armature current & Power factor

In a Synchronous motor, by varying the field excitation, it can be made to operate at different power factors

Consider a Synchronous motor having a fixed stator supply voltage & driving a constant mechanical load

Since mechanical load & speed are constant, power input to motor ($VI_a \cos\phi$) remains constant

If field excitation changes, back EMF changes

This results in a change of position of E_r & I_a . Hence power factor ($\cos\phi$) changes

The phasor diagram of Synchronous motor for different values of field excitation (Under, normal & over) are shown in figure

Under excitation

Motor is said to be under excited if $E_b < V$

Normal excitation

Motor is said to be normally excited if $E_b = V$
if $E_b = V$

Over excitation

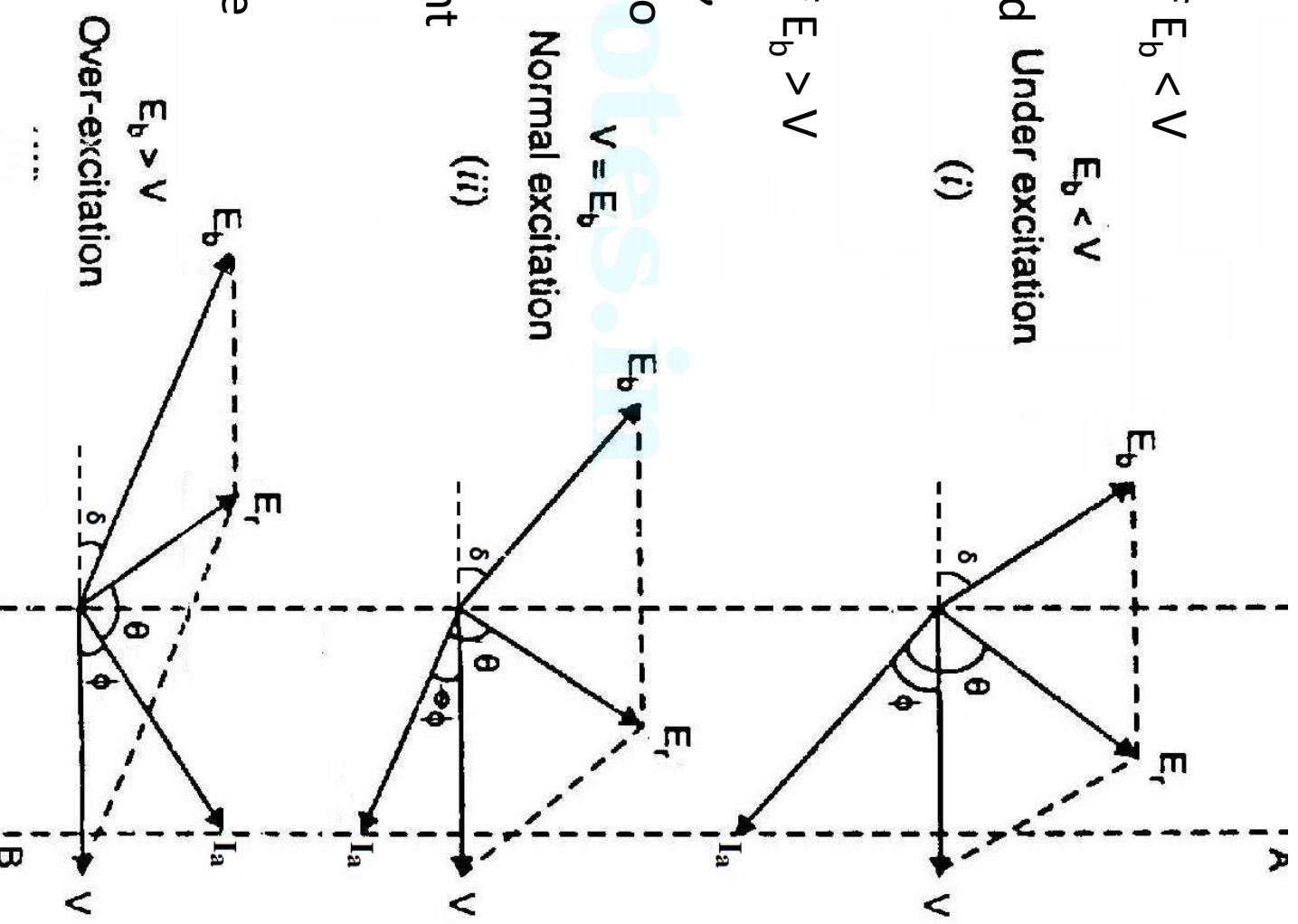
Motor is said to be over excited if $E_b > V$

From phasor diagram, it is clear that,

As excitation changes from under to over excitation, power factor changes from lagging to leading

Also magnitude of armature current changes with excitation

I_a is minimum at upf & increases in magnitude as power factor becomes poor in lag & lead directions



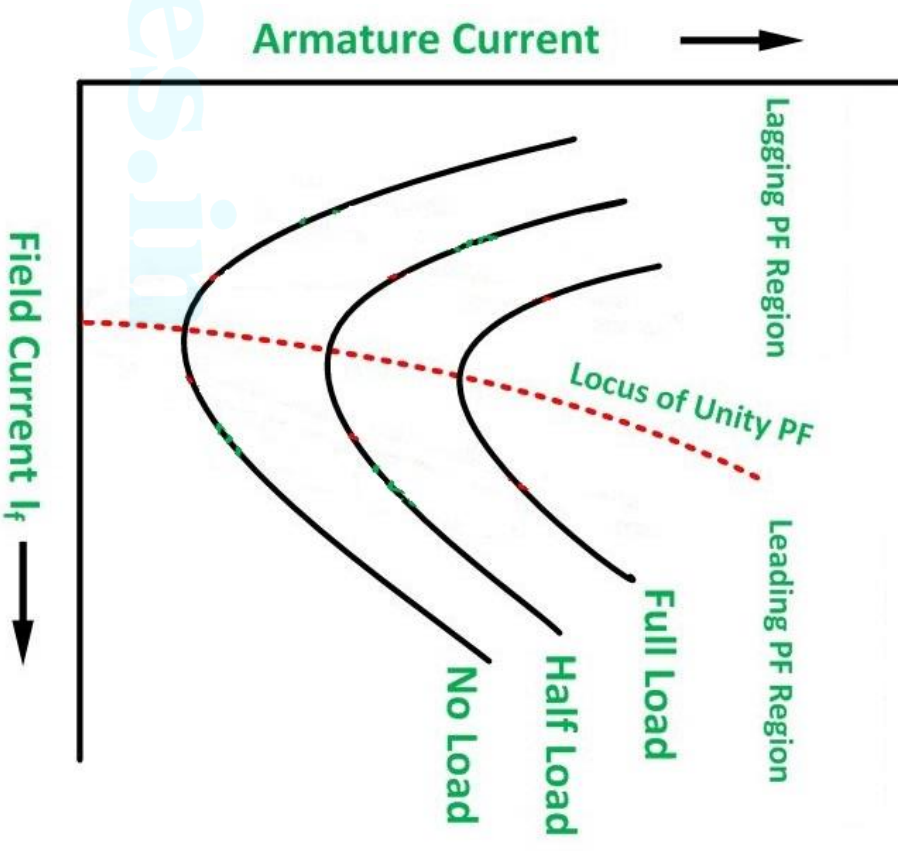
V curve

V - curve of a Synchronous motor is the graph between armature current (I_a) and field current(I_f) at constant load

In a Synchronous motor as excitation is varied, magnitude of armature current changes

I_a is minimum at upf & increases in magnitude as power factor decreases in lag & lead directions

Since it is a V-shaped curve, it is called V-curve



Inverted V – curve

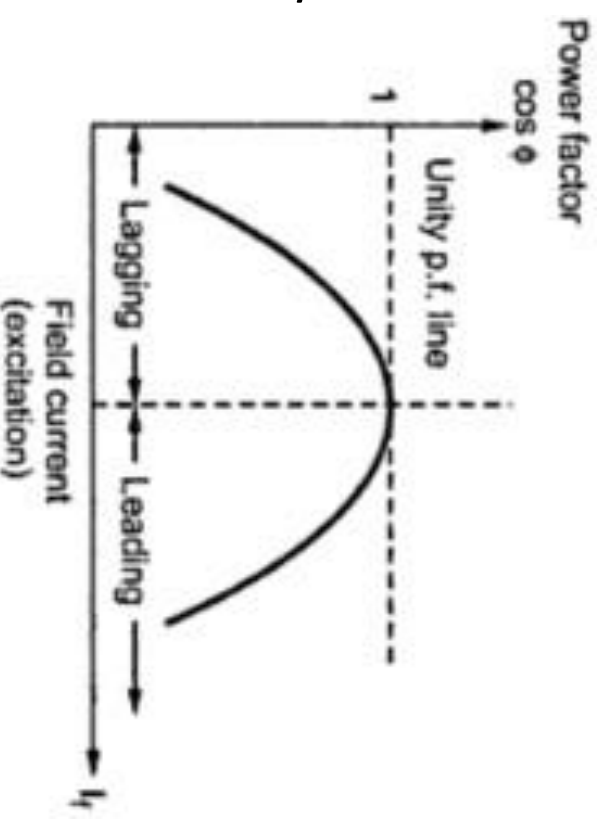
It is the graph between field current(I_f) & power factor

As field excitation changes, power factor also changes

When motor is under excited, power factor is lagging

As excitation increases, power factor increases to reach UPF

When motor is over excited, power factor increases in leading direction



Mechanical power developed by motor

Consider a Synchronous motor which is operating as under excited motor

The phasor diagram is shown in figure

The electrical power input to motor = $V I_a \cos \phi$

Since R_a is very small, stator Cu loss is small & Electrical Power input to motor = Mechanical power developed in rotor (P_m)

i.e, $P_m/\text{phase} = V I_a \cos\phi \dots (1)$

From phasor diagram,

$$AB = E_r \cos\phi$$

$$= I_a X_s \cos\phi \quad (\text{Since } I_a = E_r/X_s) \dots (2)$$

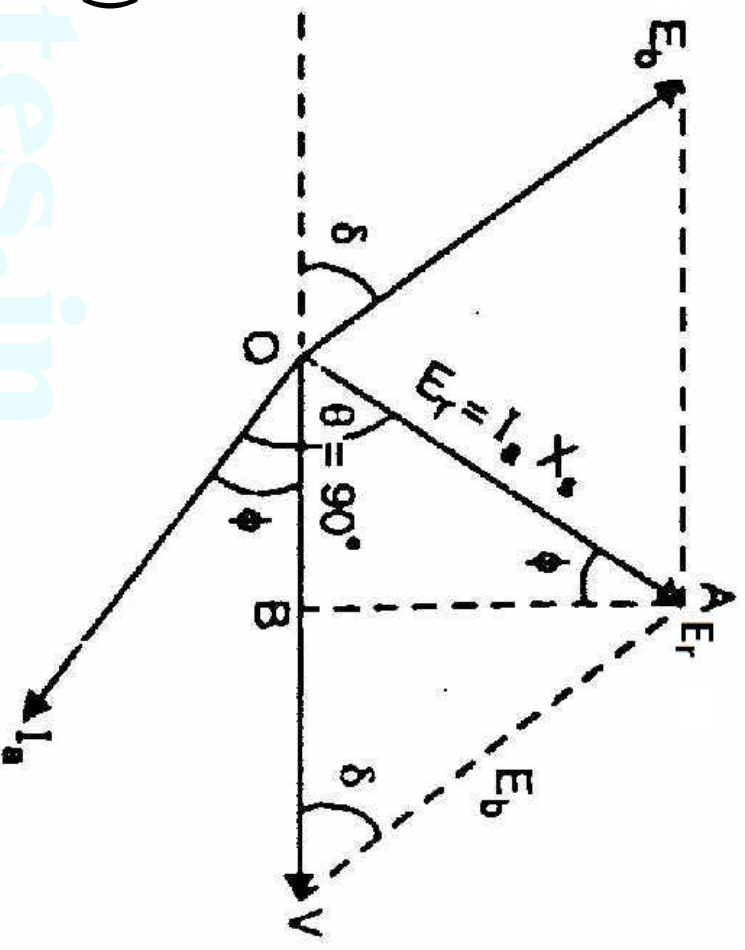
$$\text{Also, } AB = E_b \sin\delta \dots (3)$$

$$\text{Equating (2) \& (3), } \Rightarrow I_a \cos\phi = (E_b/X_s) \sin\delta$$

$$\text{Substituting in (1), } P_m = V(E_b/X_s) \sin\delta / \text{phase}$$

$$\text{3 phase Power} = 3 P_m = 3V(E_b/X_s) \sin\delta$$

From above equation, it is clear that mechanical power developed increases as angle δ increases & maximum value of δ is 90°



Power flow in a

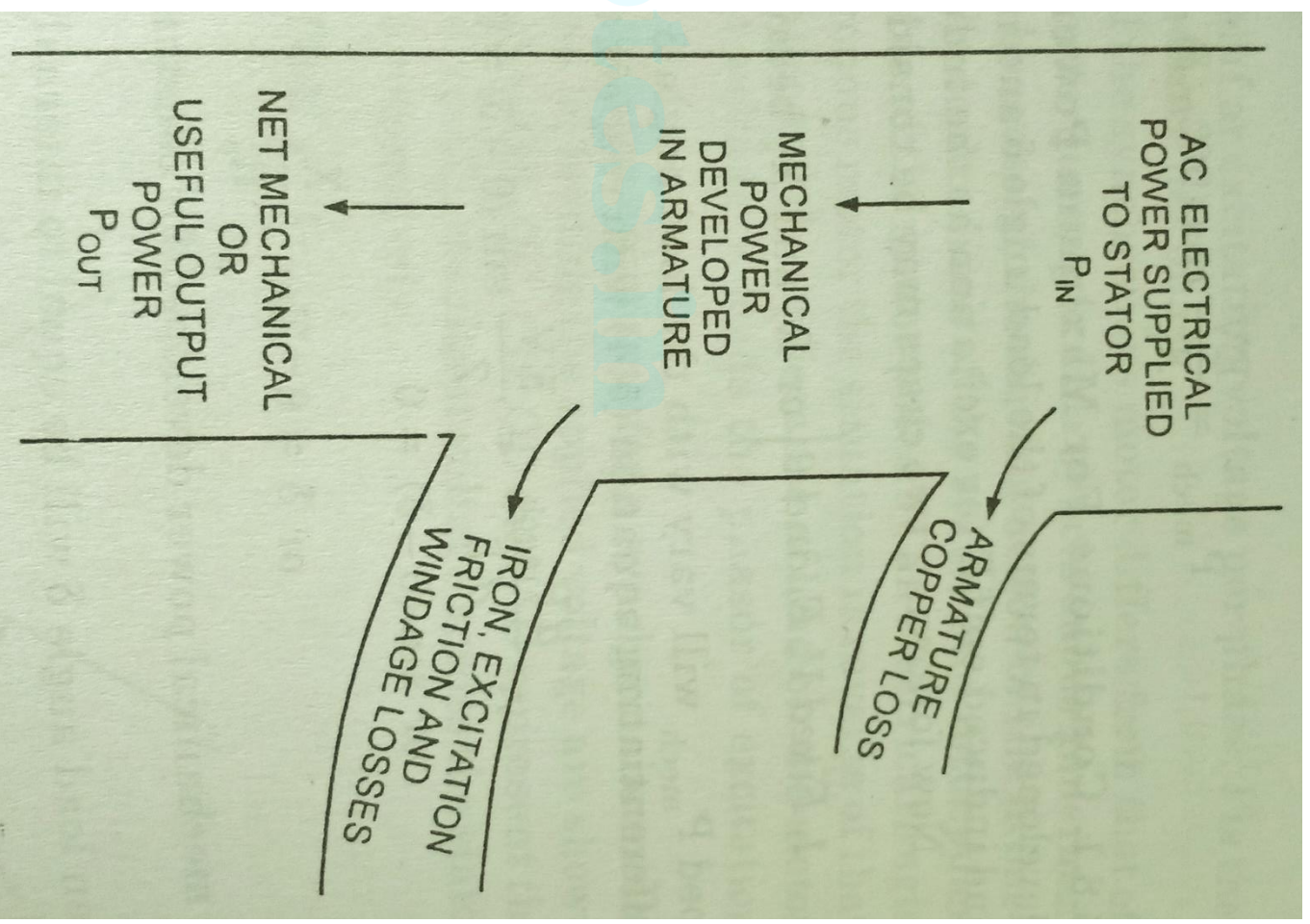
Synchronous motor

- AC electrical power from a source is given as input to motor stator
- Some of this supplied energy is utilised to overcome the armature copper loss

Remaining energy is converted into mechanical energy by motor action

Some of the mechanical energy produced is lost through friction, iron & windage loss

The remaining mechanical energy is available for useful work



Losses & Efficiency of Synchronous motor

Losses in synchronous motor may be divided into two,

a) **Load losses** – present under loaded condition, mainly due to armature current. The main load loss is armature Cu loss

Armature Cu loss – Loss occur in armature winding due to resistance of armature winding, given by $I_a^2 R_a$. It varies with load on the motor

b) **Open Circuit losses** – These losses present at no load condition. The open circuit losses are,

1. Frictional losses – losses occurring in the machine due to friction.

- Main frictional losses are bearing friction loss & brush friction loss

- These losses are small in magnitude

2. Windage loss – Losses due to air friction on rotor

3. Iron losses – These are Hysteresis & Eddy current losses in stator & rotor cores due to varying magnetic flux in the machine

- To reduce Eddy current & Hysteresis losses, core is made up of laminated Silicone steel

$$\text{Efficiency} = \frac{\text{Mechanical Power Output of Motor}}{\text{Electrical Power Input to Stator}} * 100\%$$

$$\text{Efficiency} = \frac{\text{Electrical Power Input to Stator} - \text{Losses}}{\text{Electrical Power Input to Stator}} * 100\%$$

Different Torques in Synchronous motor

1. *Starting torque* – This is the torque produced in the motor during starting.

- Since Synchronous motor is not self starting, it has no starting torque

- Modern Synchronous motors are self starting & have starting torque

2. *Running torque* – Torque produced by motor during running condition

3. *Pull in torque* – This is the torque produced when motor is pulled into synchronism when changing from induction to synchronous motor operation

4. *Pull out torque* – Maximum load torque above which motor will be pulled out of synchronism is called pull out torque

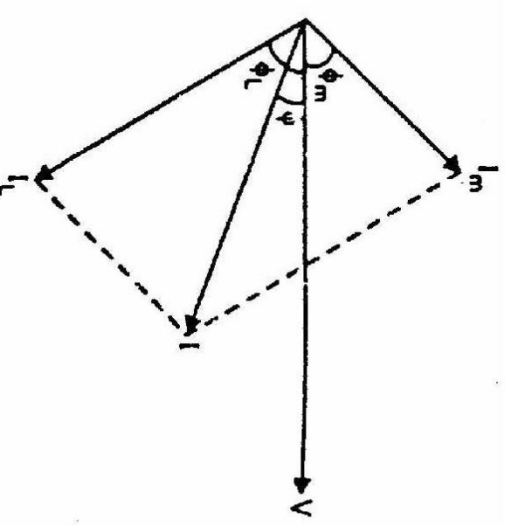
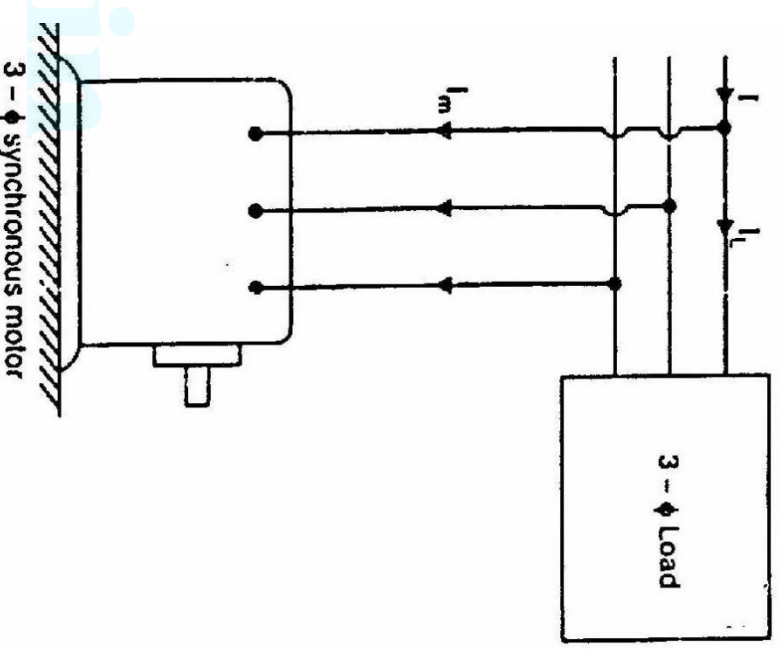
Synchronous Condenser

An over excited synchronous motor running at no load is called Synchronous condenser

Under this condition, motor will draw a leading current from supply & it can be used for power factor correction

Most of industrial loads are inductive & power factor is lagging

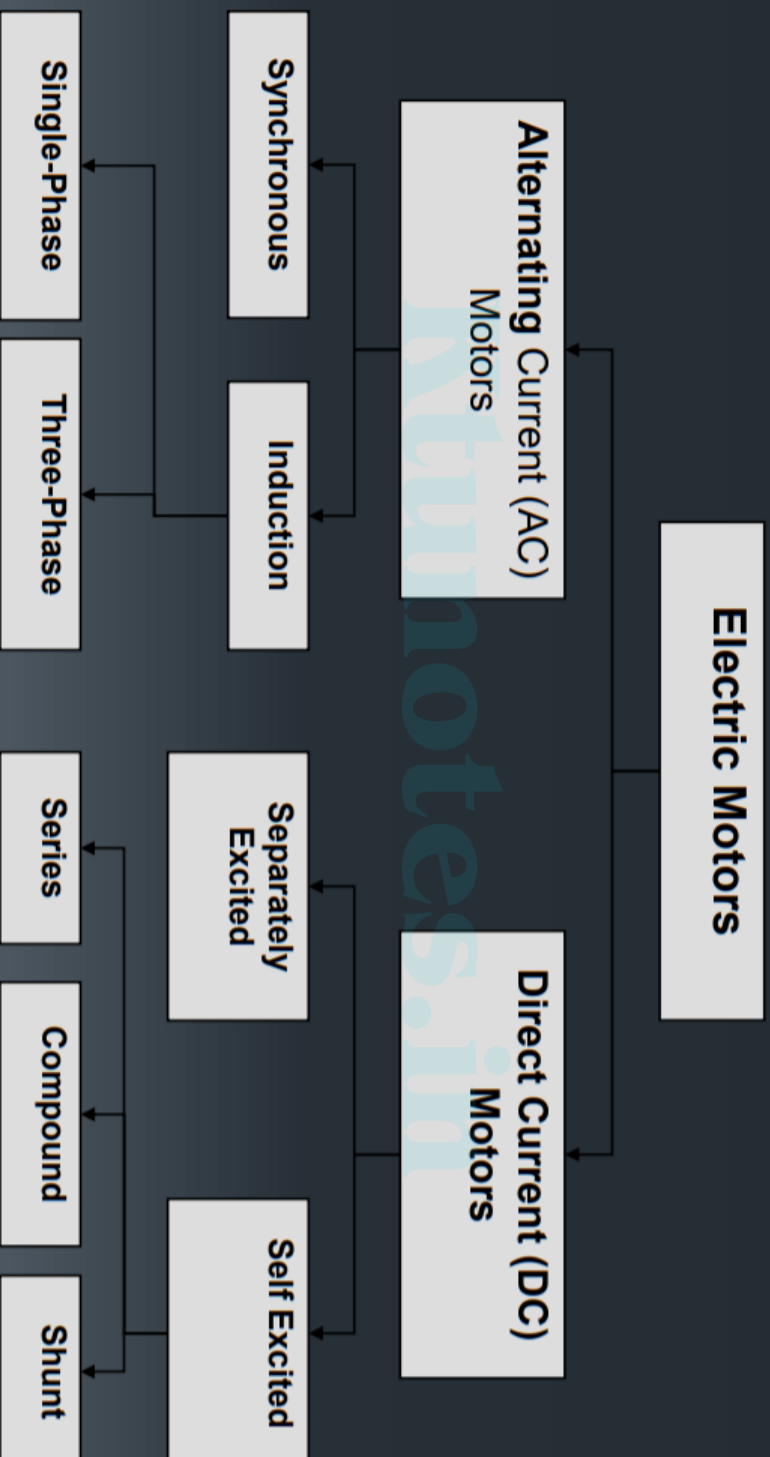
By connecting a Synchronous condenser in parallel to loads, over all power factor can be improved



INDUCTION MOTOR



CLASSIFICATION OF MOTORS



Three Phase Induction motors

Three-phase induction motors are the most common machines in industry.

More than 90% of mechanical power used in industry is supplied by 3 phase induction motors.

Three phase Induction motors are having following ***advantages***

- simple design, rugged, low-price, easy maintenance
- wide range of power ratings: fractional horsepower to 10 MW
- run essentially at constant speed from no-load to full load
- It has high efficiency and reasonably good power factor
- It has self starting torque

Disadvantages of 3 phase Induction motors are

- Speed control is difficult & requires a variable frequency power electronic drive for accurate speed control
- Starting torque is inferior to DC shunt motor

Like any electric motor, a 3 phase Induction motor has a stator and a rotor.

The stator carries a 3-phase winding (called stator winding) while the rotor carries a short-circuited winding (called rotor winding).

The stator winding is fed from 3-phase supply.

The rotor winding derives its voltage and power from the energized stator winding through electromagnetic induction and hence the name Induction motor

The induction motor may be considered to be a transformer with a rotating secondary

Construction of 3 Phase Induction Motor

An induction motor has **two main parts**

1. Stator - stationary part

2. Rotor – rotating part

- Stator is separated from rotor by a small air gap ranges from 0.4mm to 4mm, depending on the power rating of motor

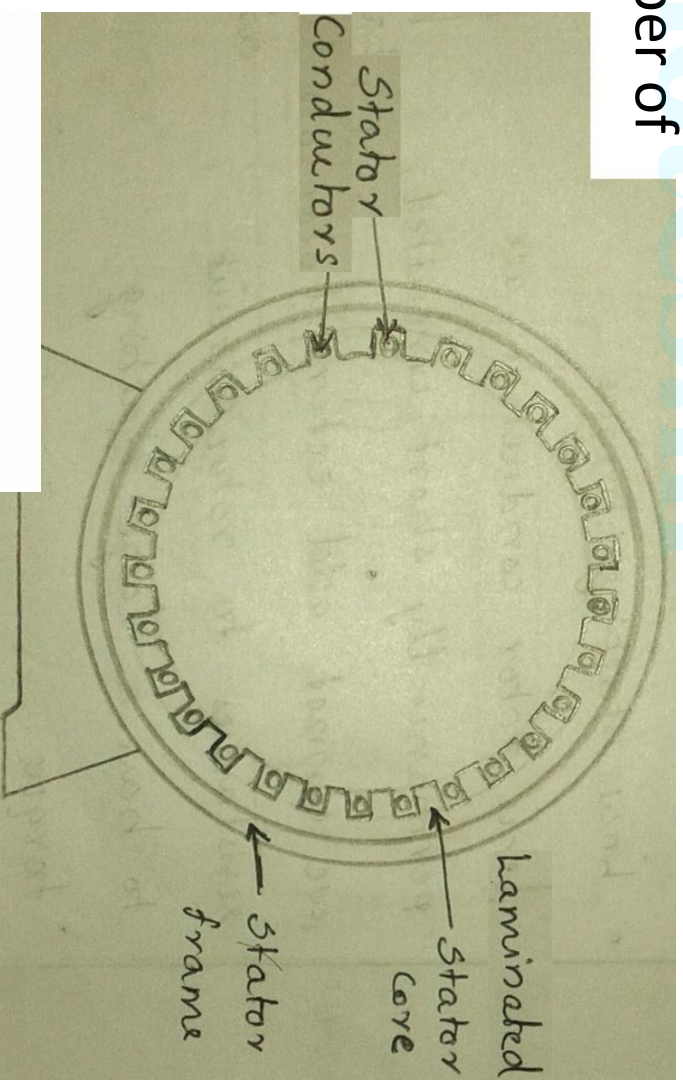
- There are two type of 3 phase Induction motor based on the construction of rotor.

- They are a) *Squirrel Cage Induction Motor (SCIM)*

b) *Slip Ring Induction Motor (SRIM)*

1. Stator -

- Consist of a steel frame that supports a hollow, cylindrical core
- Core is made of thin laminations of silicon steel – to reduce hysteresis & eddy current losses
- Core is provided with a number of evenly spaced slots in the inner periphery
- Insulated 3 phase stator windings are provided in these slots
- Windings can be star or delta connected





2. Rotor – rotating part

Depending on the construction, there are two type of rotor

a) Squirrel cage rotor

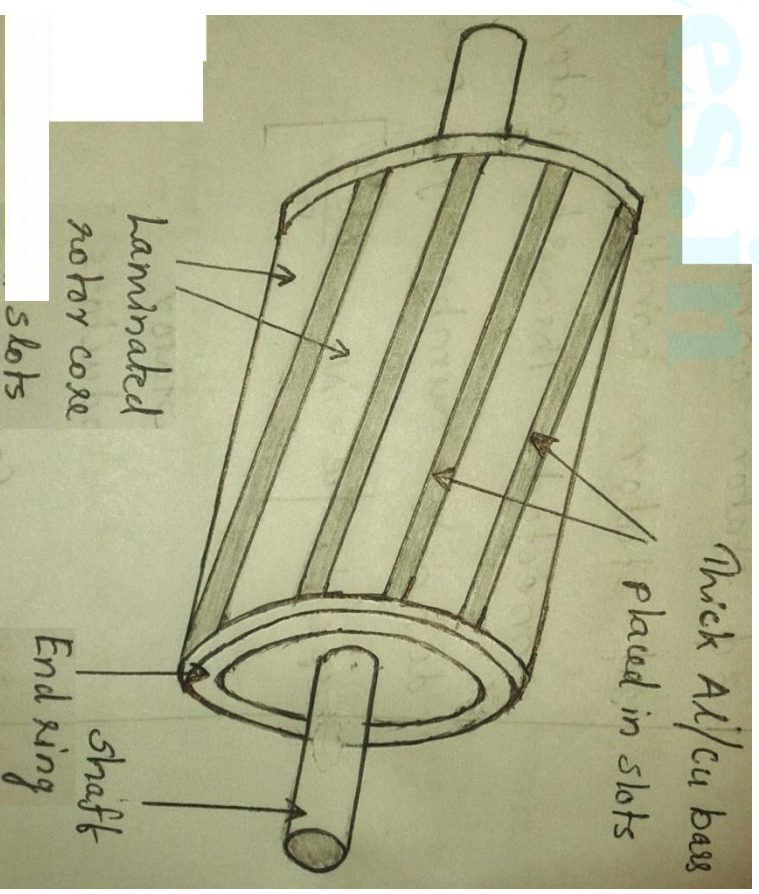
It consist of laminated cylindrical core having slots on the outer periphery

Thick copper or aluminium bars are placed in these slots

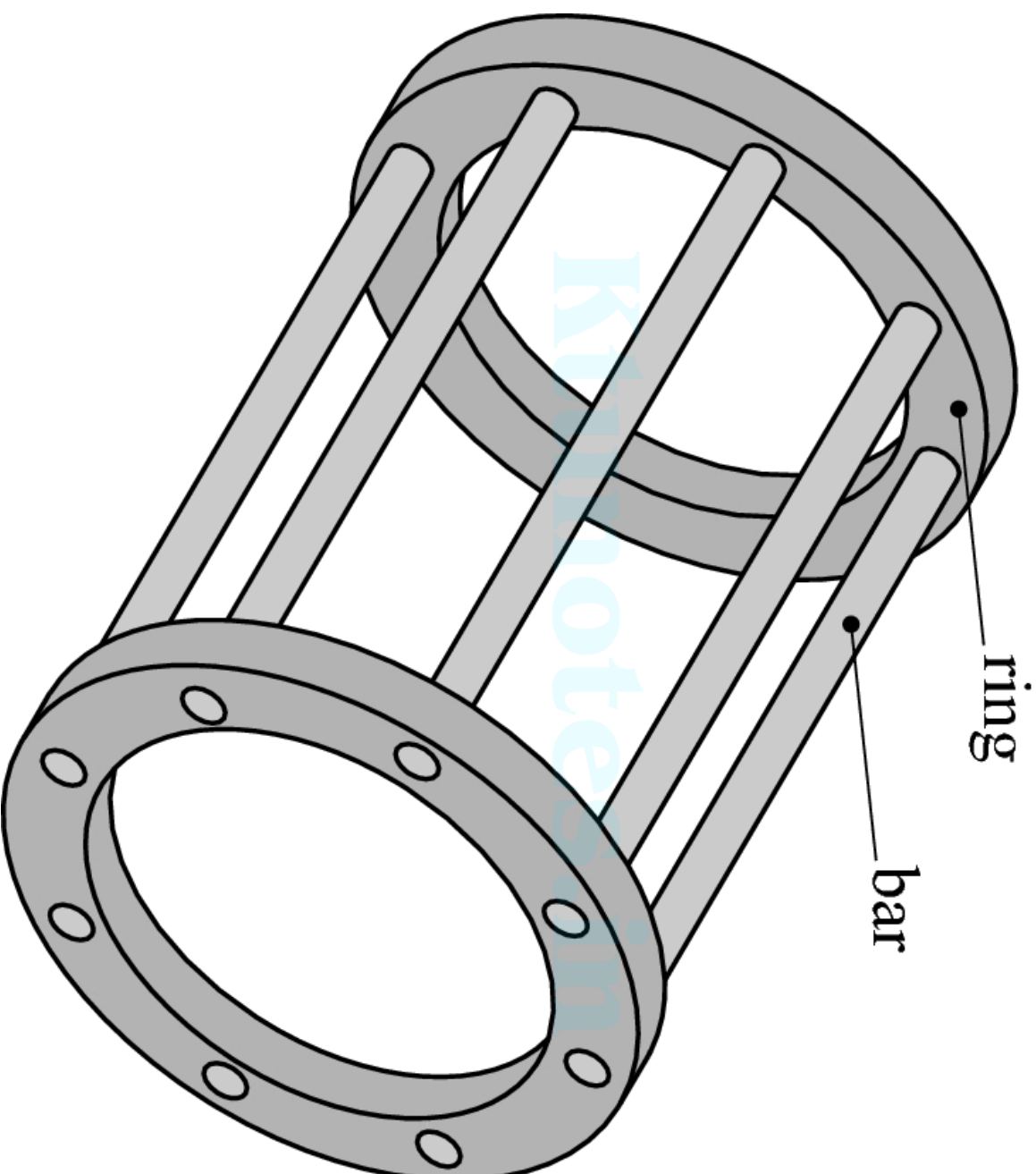
All these bars are joined together at both ends by using metal rings called end rings

Here the rotor conductors are permanently joined together and we cannot add external resistance to increase starting torque.

The entire construction (bars and end rings) resembles a squirrel cage and hence the name.



Squirrel cage rotor



b. Wound rotor

It consists of a laminated cylindrical core having slots on the outer periphery

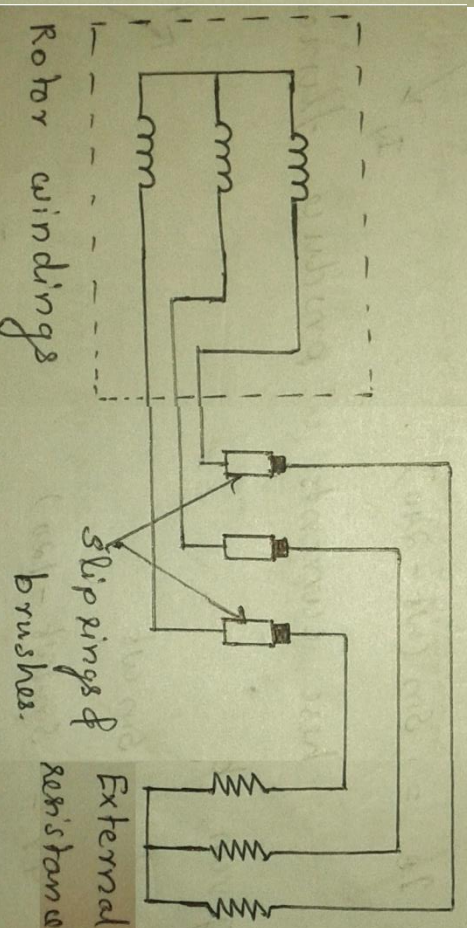
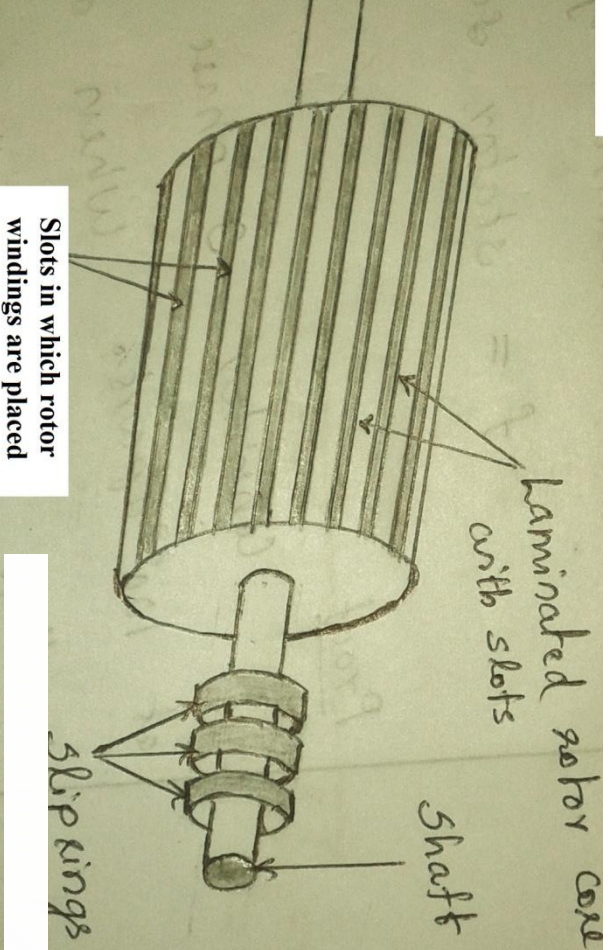
A three-phase winding is provided in these slots, similar to the winding in the stator

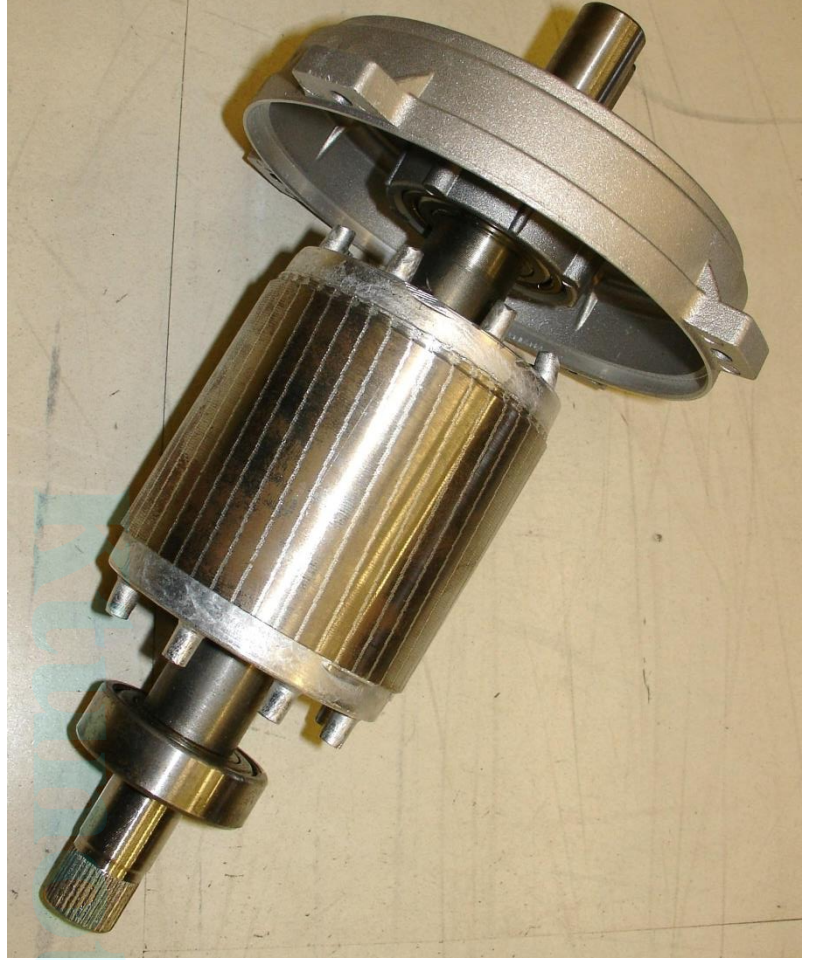
The rotor winding is usually star connected

One terminal of each rotor phase winding is brought out and connected to 3 slip rings mounted on the rotor shaft.

By using brushes, external resistances can be connected to the rotor windings to increase starting torque.

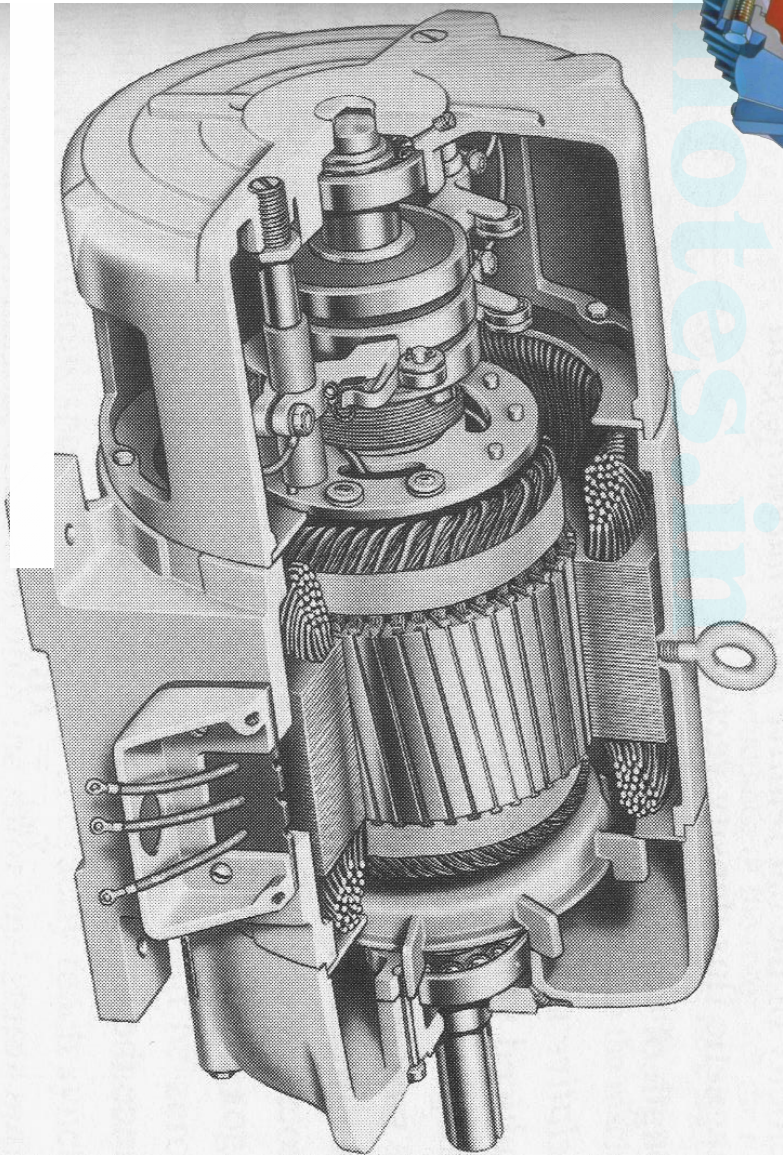
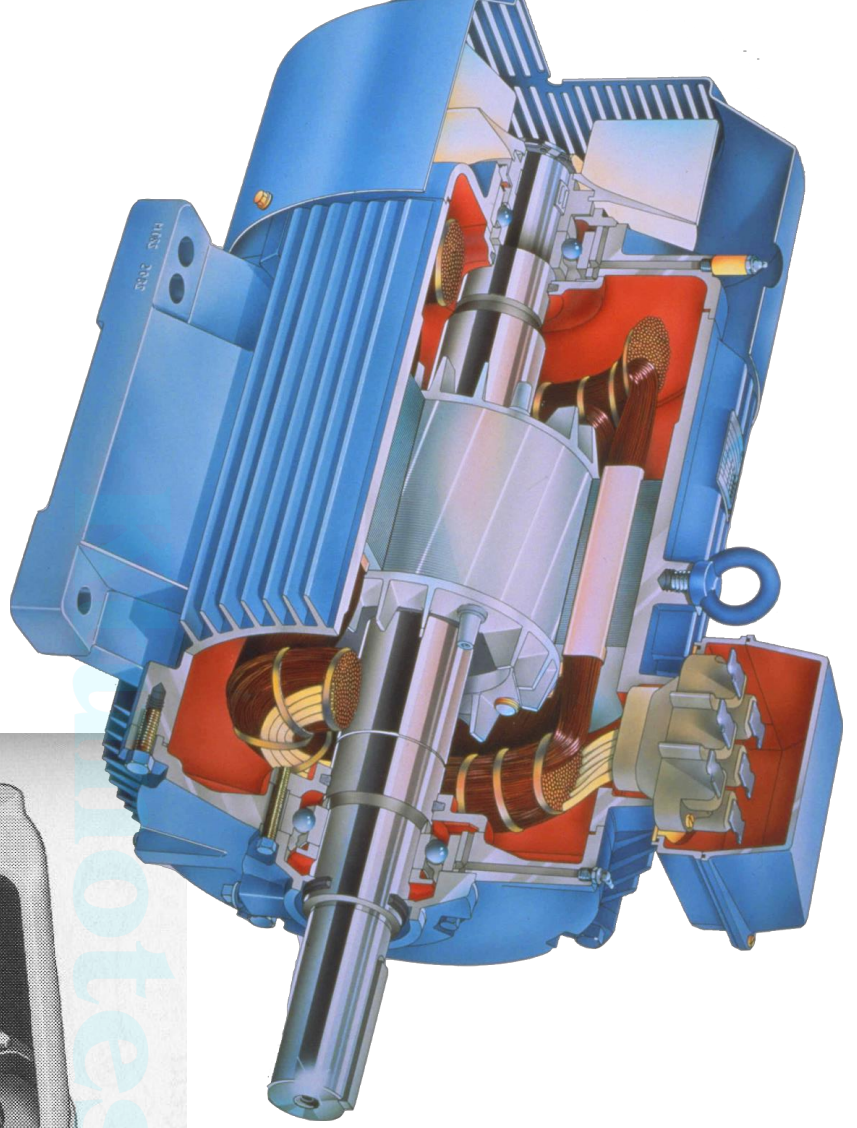
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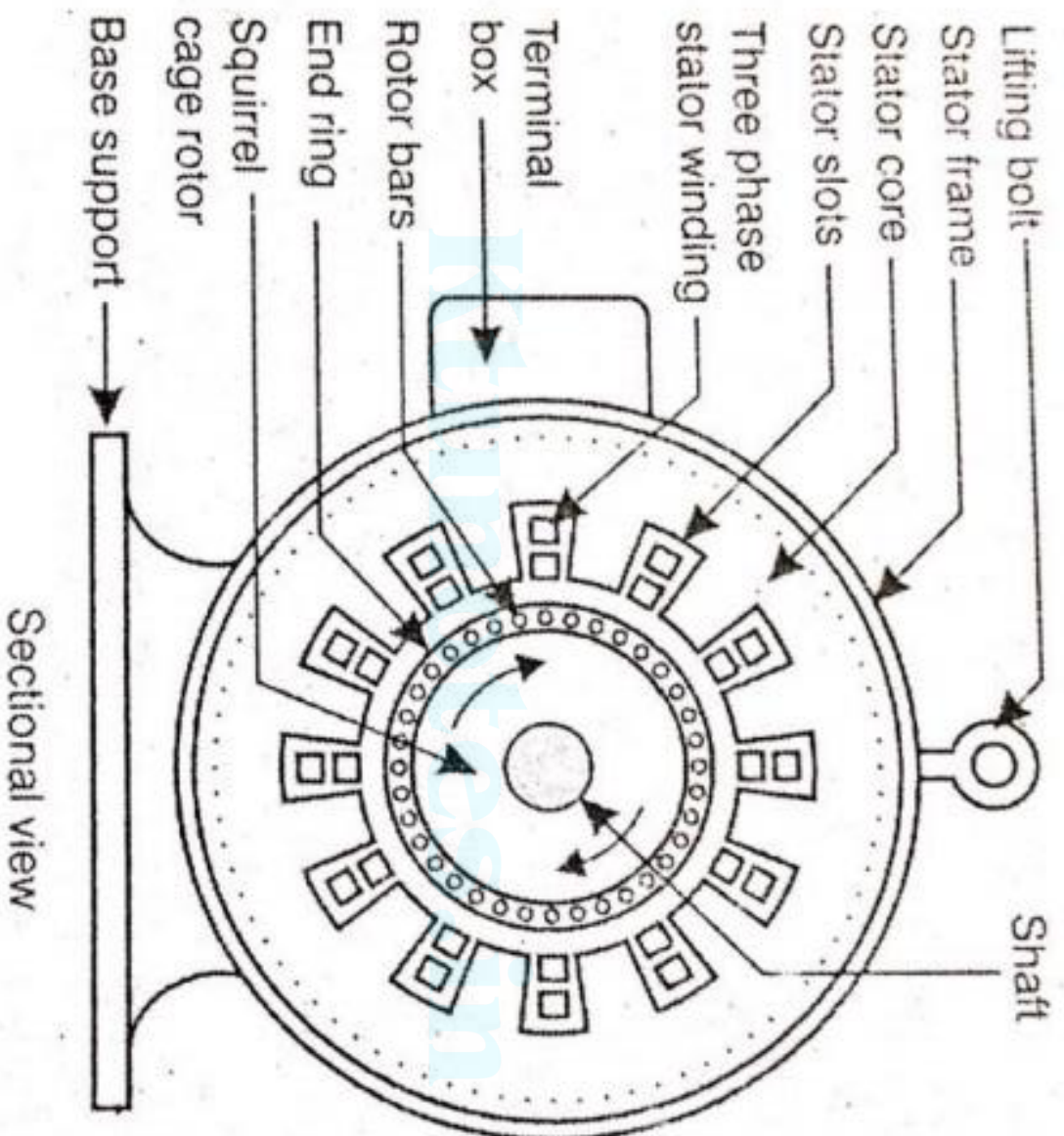
Parts of AC Motor





SCIM		SRIM	
Construction is simple and rugged .		Construction is complicated .	
Cheap		Costly	
Rotor consists of bars of conductors shorted at the end by end rings .		Rotor consists of 3 phase winding similar to stator winding.	
External resistance cannot be added		External resistance can be added to increase the starting torque.	
Used for constant speed operation .		Used where high starting torque is required.	
Requires less conductor material		Requires more conductor material	
Brushes and slip rings are absent.		Slip rings and brushes are present to add external resistance .	
Less Maintenance		Due to brushes, frequent maintenance is required.	

SCIM	SRIM
Moderate starting torque and cannot be controlled.	High starting torque with low starting current.
Rotor resistance starter cannot be used.	Rotor resistance starter can be used.
Rotor copper loss is low. So efficiency is high.	Rotor copper loss is high. So efficiency is low.
Rotor automatically adjust itself for same no. of stator poles.	Rotor is also wound for same no. of poles as that of stator.
95% of IM in industry are SCIM	Only 5% of IM in industry are SRIM.
Used in • Fans • Blowers • Drilling & Printing machines • Pumps	Used in • Lifts • Cranes • Compressors • Elevators



Rotating Magnetic Field

The principle of a 3-phase, two-pole stator having three identical windings placed 120 space degrees apart is shown in Fig. 34.11. The flux (assumed sinusoidal) due to three-phase windings is shown in Fig 34.12.

The assumed positive directions of the fluxes are shown in Fig 34.13. Let the maximum value of flux due to any one of the three phases be Φ_m . The resultant flux Φ_r , at any instant, is given by the vector sum of the individual fluxes, Φ_1 , Φ_2 and Φ_3 due to three phases. We will consider values of Φ_r at four instants 1/6th time-period apart corresponding to points marked 0, 1, 2 and 3 in Fig. 34.12

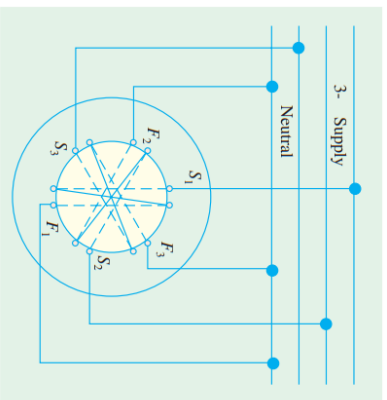


Fig. 34.11

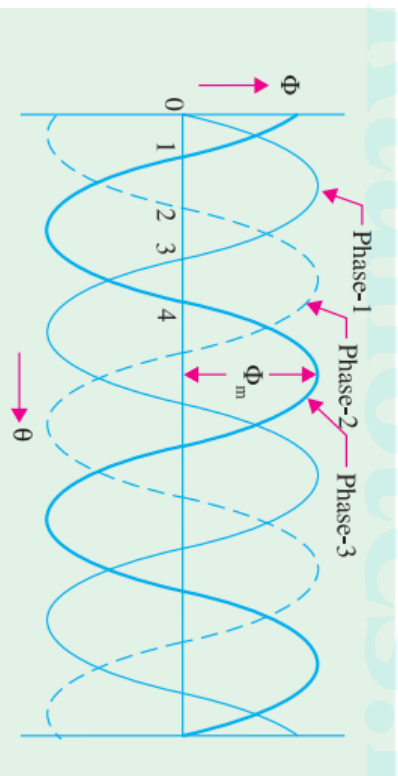


Fig. 34.12

$$\begin{aligned}\Phi_1 &= \Phi_m \sin(\omega t) \\ \Phi_2 &= \Phi_m \sin(\omega t - 120^\circ) \\ \Phi_3 &= \Phi_m \sin(\omega t - 240^\circ)\end{aligned}$$

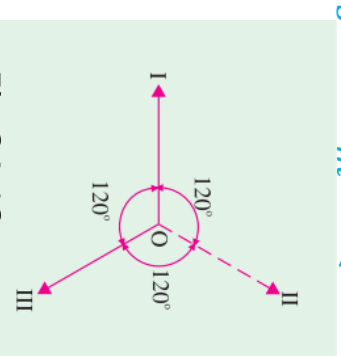


Fig.34.13

(ii) when $\theta = 60^\circ$ i.e. corresponding to point 1 in Fig. 34.12.

$$\text{Here } \Phi_1 = \frac{\sqrt{3}}{2} \Phi_m$$

...drawn parallel to OI of Fig. 34.13 as shown in Fig. 34.14 (ii)

$$\Phi_2 = -\frac{\sqrt{3}}{2} \Phi_m$$

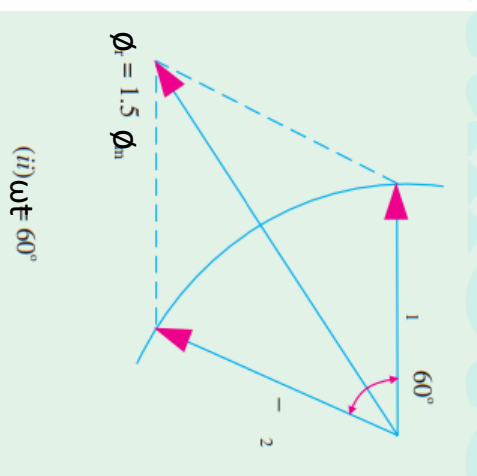
...drawn in opposition to OII of Fig. 34.13.

$$\Phi_3 = 0$$

$$\therefore \Phi_r = 2 \times \frac{\sqrt{3}}{2} \Phi_m \times \cos 30^\circ = \frac{3}{2} \Phi_m$$

[Fig. 34.14 (ii)]

It is found that the resultant flux is again $\frac{3}{2} \Phi_m$ but has rotated clockwise through an angle of 60° .

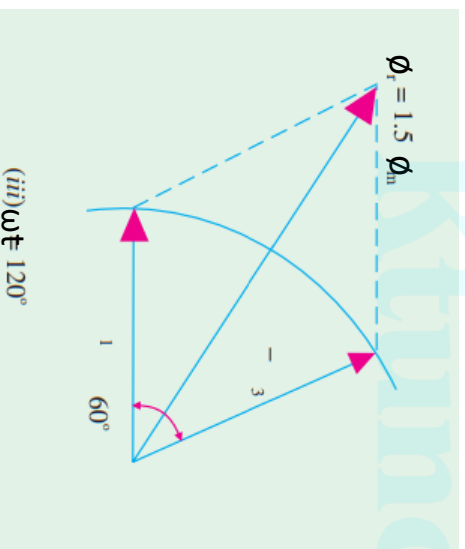


(iii) When $\theta = 120^\circ$ i.e. corresponding to point 2 in Fig. 34.12.

$$\text{Here, } \Phi_1 = \frac{\sqrt{3}}{2} \Phi_m, \quad \Phi_2 = 0, \quad \Phi_3 = -\frac{\sqrt{3}}{2} \Phi_m$$

It can be again proved that $\Phi_r = \frac{3}{2} \Phi_m$

So, the resultant is again of the same value, but has further rotated clockwise through an angle of 60° [Fig. 34.14 (iii)].

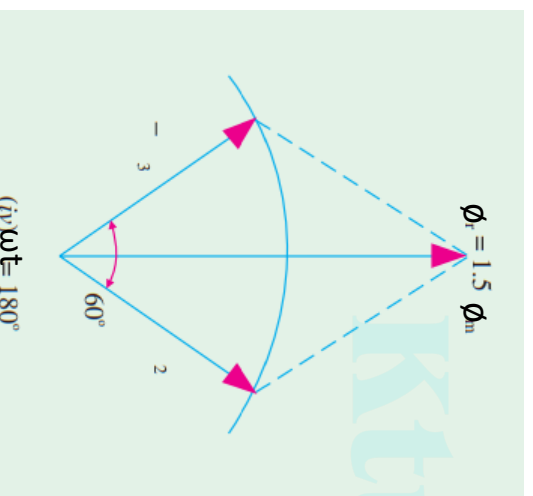


(iv) When

$\theta = 180^\circ$ i.e. corresponding to point 3 in Fig. 34.12.

$$\Phi_1 = 0, \Phi_2 = \frac{\sqrt{3}}{2} \Phi_m, \Phi_3 = -\frac{\sqrt{3}}{2} \Phi_m$$

The resultant is $\frac{3}{2} \Phi_m$ and has rotated clockwise through an additional angle 60° or through an angle of 180° from the start.



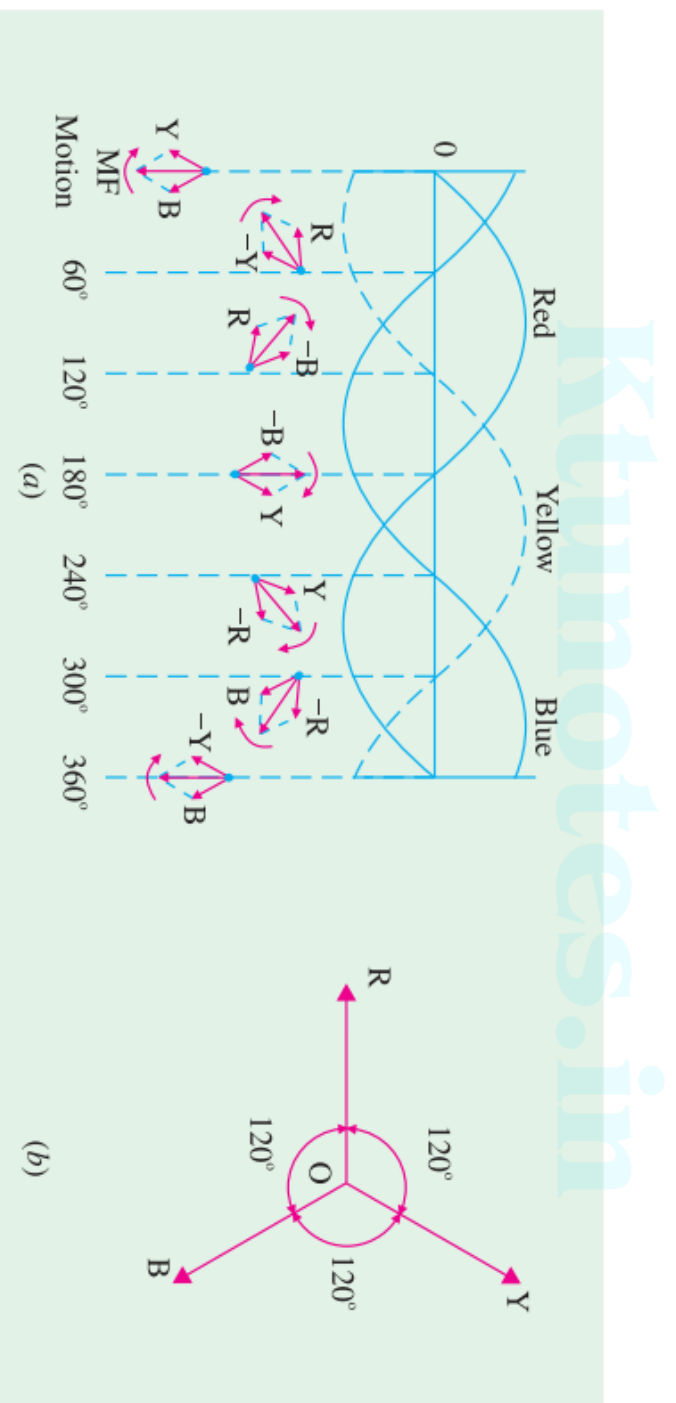
Hence, we conclude that

1. the resultant flux is of constant value = $1.5 \Phi_m$ i.e. 1.5 times the maximum value of the flux due to any phase.

2. the resultant flux rotates around the stator at synchronous speed given by $N_s = \frac{120}{P} f/P$.

Where F = frequency of applied voltage

And P = No of poles



Principle of operation

When 3 phase supply is given to 3 phase stator winding, a rotating magnetic field is produced

This rotating magnetic field links the stationary rotor windings and produces an induced emf in the rotor windings

Since the rotor windings are short circuited for both squirrel cage and wound-rotor, a large current flows through the rotor windings. Now the situation is exactly like *a current carrying conductor placed in a magnetic field*

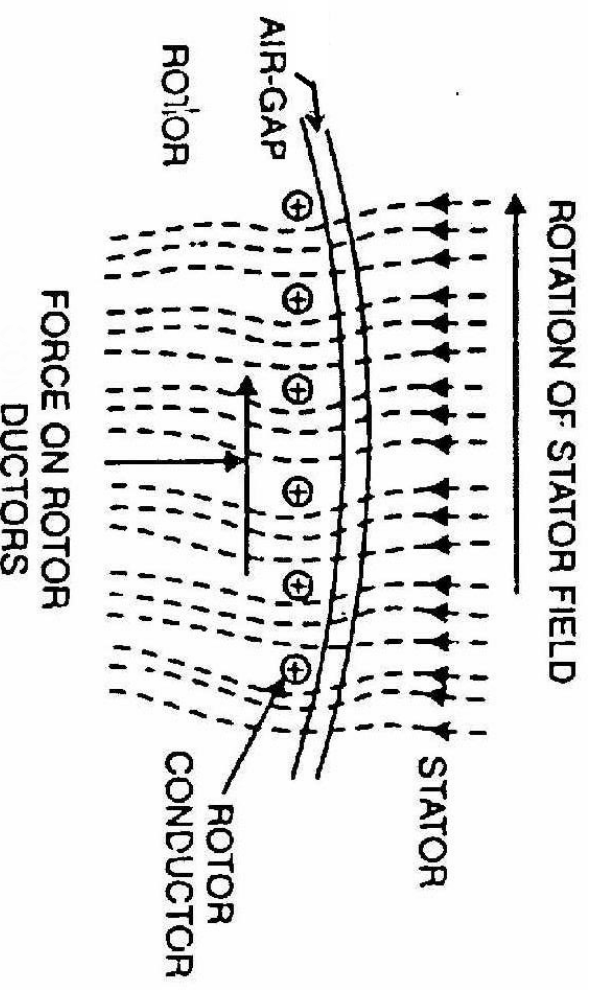
The current carrying conductor placed in a magnetic field experiences a mechanical force/torque

Thus rotor conductors experiences a force/torque, and it tends to rotate the rotor in the same direction of rotating magnetic field (Lenz's Law)

i.e, the direction of rotor current will be such as to oppose the cause producing it .

i.e, the relative speed between rotating magnetic field and rotor conductors

Hence to reduce the relative speed, the rotor starts rotating in the same direction of stator field and tries to attain the same speed of rotating magnetic field



Induction motor speed

- Can the Induction Motor run at the synchronous speed??
- If rotor runs at the synchronous speed, which is the same speed of the rotating magnetic field, then the rotor will appear stationary to the rotating magnetic field and the rotating magnetic field will not cut the rotor conductors. So, no induced current will flow in the rotor and net torque generated is zero. Hence the rotor speed will fall below the synchronous speed due to load torque and inertia of rotor
- When the speed falls, the rotor windings will cut the rotating magnetic field and a torque is produced and rotor again accelerates
- Due to inertia of rotor, motor will run at a speed which is less than synchronous speed
- The difference between the actual motor speed and the synchronous speed is called the *Slip speed (N_{slip})*

$$N_{slip} = N_s - N$$

Where N_{slip} = slip speed

N_s = speed of the rotating magnetic field

N = shaft speed of the motor

Slip

The difference between the synchronous speed(N_s) and the actual speed of rotor(N) is called **slip(S)**

Expressed as the % of synchronous speed

$$S = \frac{N_s - N}{N_s} * 100\%$$

Notice that : if the rotor runs at synchronous speed, $S = 0$

if the rotor is stationary, $S = 1$

Rotor current Frequency

The frequency of the voltage/current induced in the rotor is given by

$$f_r = \frac{N_{slip} P}{120} = S f$$

.e, rotor current frequency = slip * stator supply frequency

1. A 3-φ induction motor is wound for 4 poles and is supplied from 50-Hz system. Calculate (i) the synchronous speed (ii) the rotor speed, when slip is 4% and (iii) rotor frequency when rotor runs at 600 rpm.

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Solution. (i)

$$N_s = 120 f/P = 120 \times 50/4 = 1500 \text{ rpm}$$

(ii) rotor speed,

$$N = N_s (1 - s) = 1500 (1 - 0.04) = \mathbf{1440 \text{ rpm}}$$

(iii) when rotor speed is 600 rpm, slip is

$$s = (N_s - N)/N_s = (1500 - 600)/1500 = 0.6$$

rotor current frequency,

$$f' = sf = 0.6 \times 50 = \mathbf{30 \text{ Hz}}$$

Rotor EMF(E2)

At standstill condition, 3-phase induction motor is equivalent to 3-phase transformer with secondary short-circuited.

∴ rotor induced emf per phase(E2), at standstill condition,

$$E_2 = E_1 \times \frac{N_2}{N_1}$$

where E_1 = Applied voltage per phase to stator,

N_2 = Number of turns per phase on rotor,

N_1 = Number of turns per phase on stator

When rotor rotates, the relative speed of rotor with respect to stator field drops in proportion to slip(s).

∴ **Rotor induced emf = $s \times E_2$**

ie, for slip s, induced emf in rotor is 's' times the induced emf in

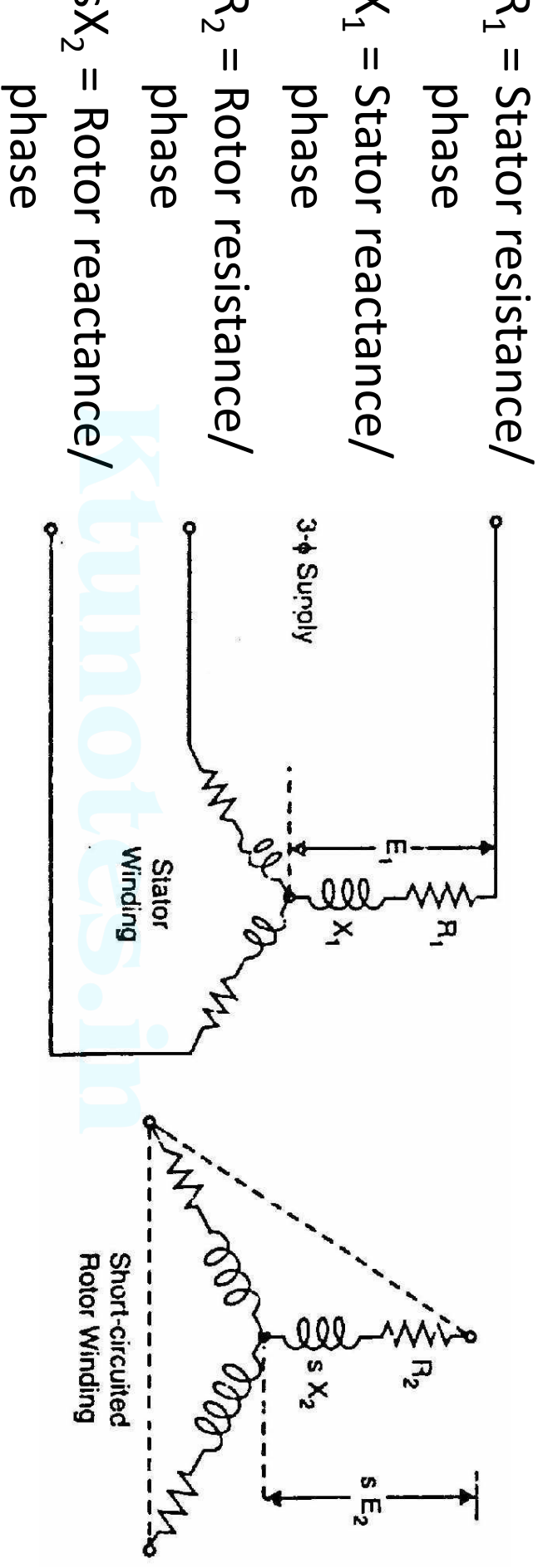
-the rotor at standstill condition.

-Rotor frequency = Sf

-Rotor reactance/phase = SX

Rotor Current

The circuit of a 3 phase SRIM is shown in figure



E_1 = Stator EMF/phase

sE_2 = Rotor EMF/phase

3 phase Induction motor is considered as a balanced load. So we need to consider only one phase.

At standstill

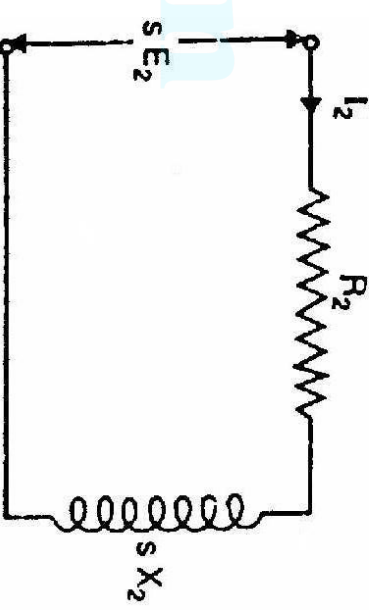
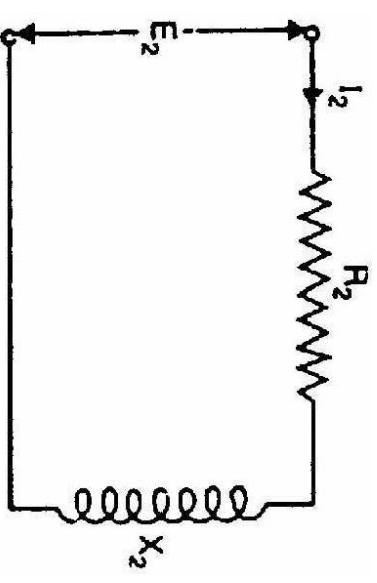
One phase of rotor under standstill condition (S=1) is shown in figure.

$$\text{Rotor Current / phase, } I_2 = \frac{E_2}{Z_2} = -\frac{E_2}{\sqrt{R_2^2 + X_2^2}}$$

$$\text{Rotor Power factor, } \cos \phi_2 = \frac{R_2}{Z_2} = -\frac{R_2}{\sqrt{R_2^2 + X_2^2}}$$

When running at Slip S

One phase of rotor under this condition is shown in figure.



$$\text{Rotor Current / phase, } I_2 = \frac{sE_2}{Z_2} = -\frac{sE_2}{\sqrt{R_2^2 + s^2 X_2^2}}$$

$$\text{Rotor Power factor, } \cos \phi_2 = \frac{R_2}{Z_2} = -\frac{R_2}{\sqrt{R_2^2 + s^2 X_2^2}}$$

Rotor Torque

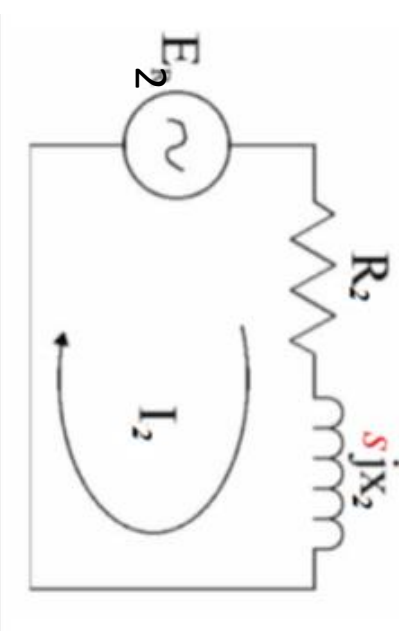
• Torque produced in rotor depends on rotor current, rotor power factor & stator field

i.e, $T \propto \phi I_2 \cos \phi_2$

$T \propto E_2 I_2 \cos \phi_2$ (Since $\phi \propto E_2$)

$T = K E_2 I_2 \cos \phi_2 \dots (1)$

Torque under running condition



We have

Sub I_2 & $\cos \phi_2$ in equation (1)

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + s^2 X_2^2}}$$

$$\cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + s^2 X_2^2}}$$

T

$$= K E_2 \frac{sE_2}{\sqrt{R_2^2 + s^2 X_2^2}} \frac{R_2}{\sqrt{R_2^2 + s^2 X_2^2}}$$

$$T = \frac{K_s R_2 E_2^2}{R_2^2 + (sX_2)^2} \quad \dots (2)$$

Starting torque (T_{st})

The torque produced by motor during starting (when $S = 1$) is called starting torque

At starting, $S = 1$. Substituting in equation (2)

$$T_{st} = \frac{K R_2 E_2^2}{R_2^2 + X_2^2}$$

$E_2 \propto V$, supply voltage & is a constant

$$\therefore T_{st} = \frac{K_1 R_2}{R_2^2 + X_2^2}$$

Condition for maximum starting torque

Starting torque can be varied by varying rotor resistance, R_2

To get condition, take derivative of T_{st} w.r.to R_2 & equate to 0

$$\frac{d(T_{st})}{dR_2} = 0 \quad \Rightarrow \mathbf{R_2 = X_2}$$

$\mathbf{R_2 = X_2}$ is the condition for maximum starting torque

Pull out torque/Maximum torque/Breakdown torque

The maximum torque that can be produced by an Induction motor is called pull out torque

Torque produced by motor is given by, $T = \frac{KsR_2E_2^2}{R_2^2 + (sX_2)^2}$

Supply voltage is constant, E_2 will be a constant. Therefore,

$$T \propto \frac{sK_2}{R_2^2 + (sX_2)^2}$$

By differentiating T wrt s we get the condition for maximum torque as $R_2 = sX_2$

On substituting the condition in torque equation, we get maximum torque as,

$$T_{\max} = \frac{KE_2^2}{2X_2}$$

i.e, maximum torque is independent of R_2 ,

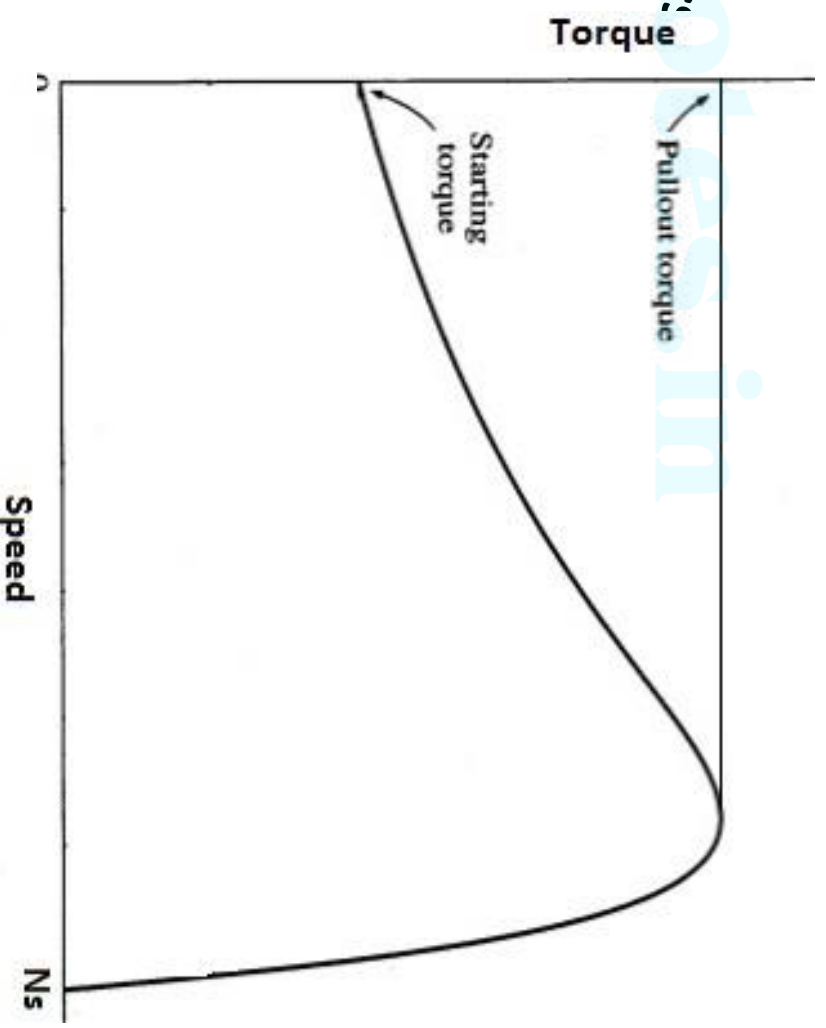
As slip increases, speed decreases & torque increases. Torque reaches maximum value at $s = (R_2/X_2)$. Maximum torque is known as breakdown torque or pullout torque.

With higher value of slip, R_2 become negligible compared to sX_2 &

$T \propto (s/X_2)^2 \propto (1/s)$, i.e, curve is a hyperbola.

With further increase in slip, torque decreases, motor slows down & stops.

The Torque-Speed characteristic of an Induction motor is shown in figure



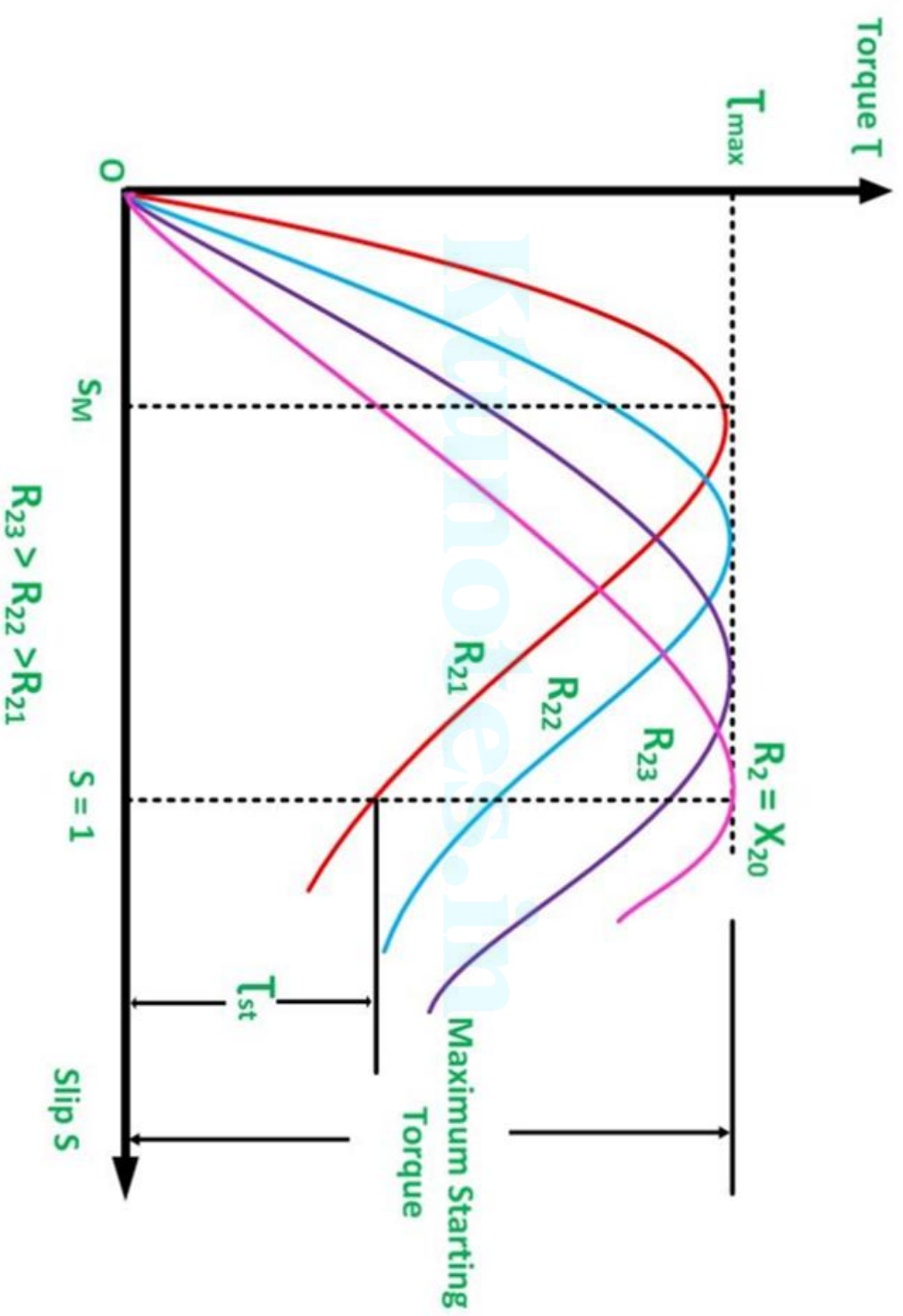
Full load torque (T_f)

- The torque produced by motor when it takes full load current (rated current) is called full load torque
- Let full load slip = S_f , now

$$T_f = \frac{K S_f R_2 E_2^2}{R_2^2 + (s_f X_2)^2}$$

Effect of varying Rotor resistance

- Maximum torque remains constant.
- The Value of S_m depends on the value of R_2 .
- Greater the R_2 , greater is the value of slip at which the maximum torque occurs.
- Starting torque increases.
- Reduces the starting current.
- Improves the PF at starting.



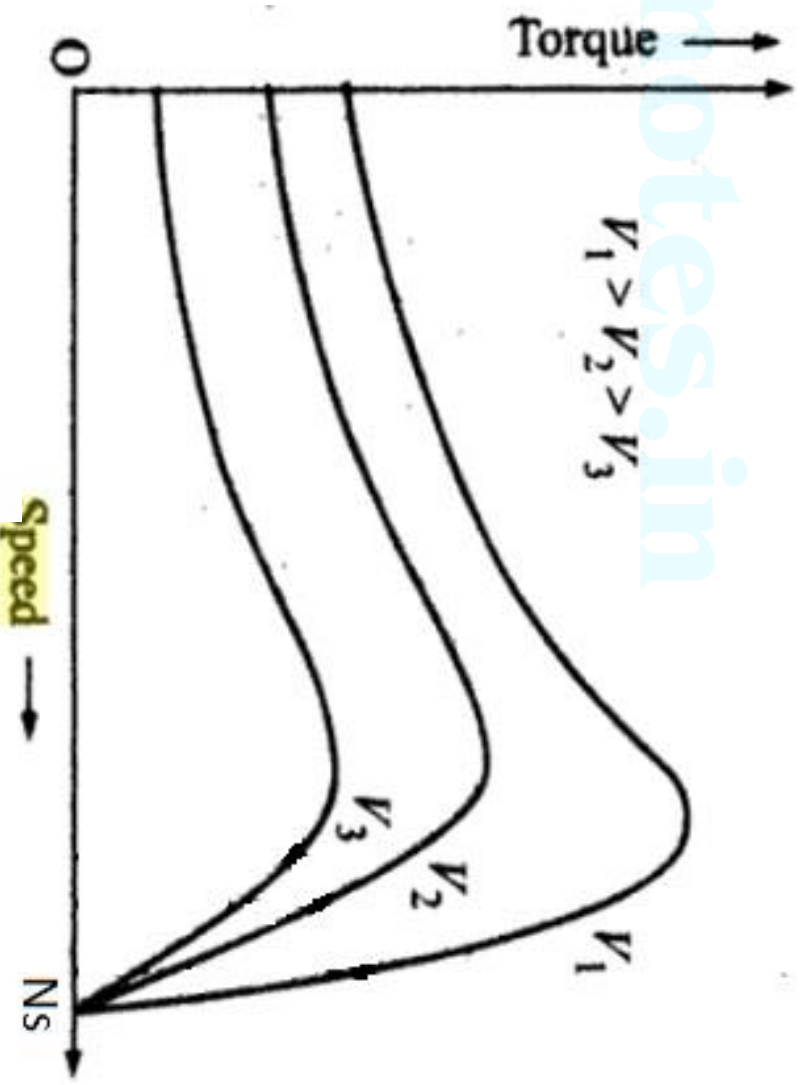
Effect of varying Stator supply voltage on T – Speed

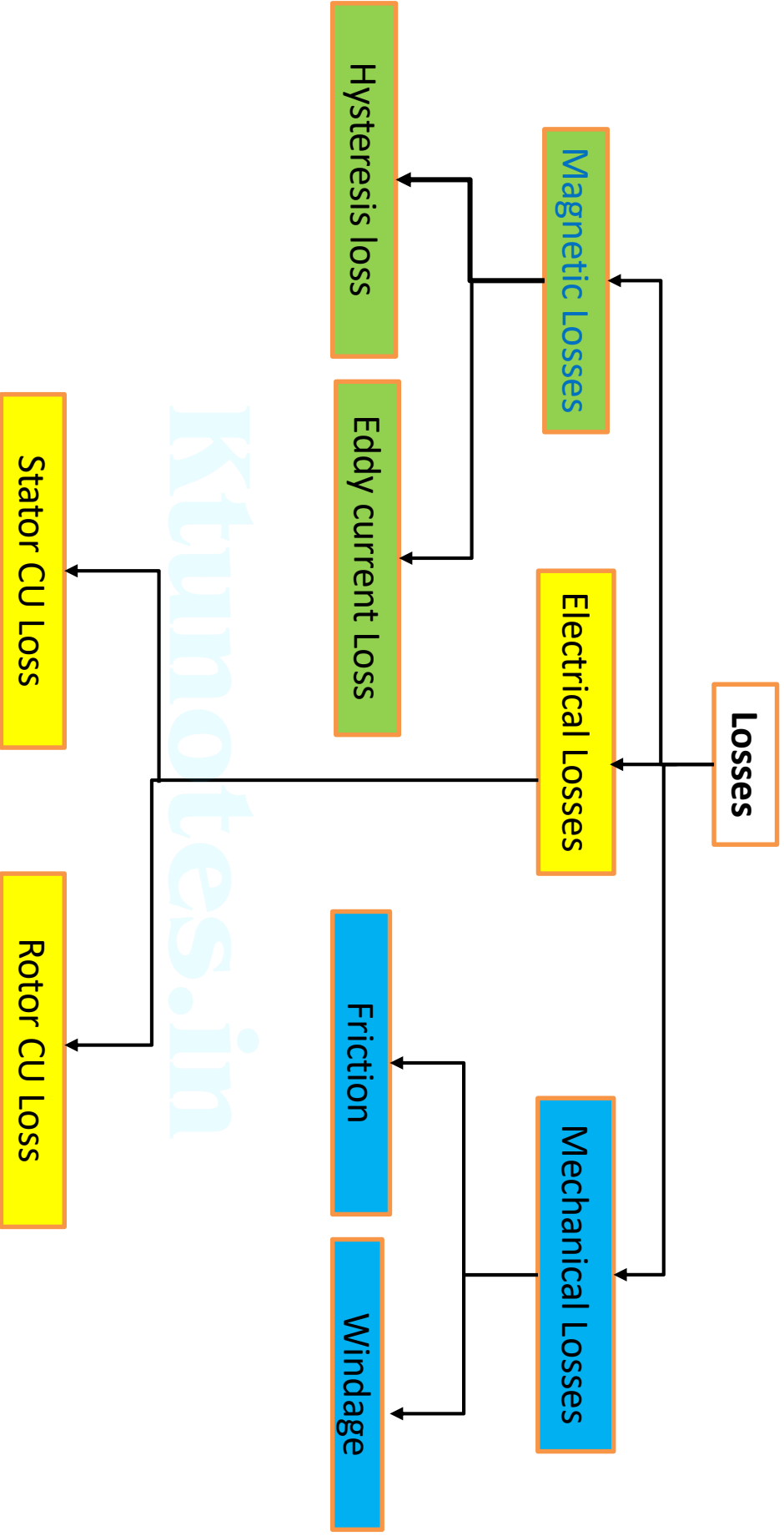
characteristics

$T \propto E_2^2 \propto V_2^2$ (Since, $E_2 \propto V$, stator voltage)

Also $T_{st} \propto E_2^2 \propto V_2^2$

The variation of Torque with stator voltage is shown in figure





Hysteresis loss is due to the reversal of magnetization of the armature core. When the core passes under one pair of poles, it undergoes one complete cycle of magnetic reversal. The frequency of magnetic reversal is given by, $f = P \cdot N / 120$ (where, P = no. of poles and N = Speed in rpm)

The loss depends upon the volume and grade of the iron, frequency of magnetic reversals and value of flux density. Hysteresis loss is given by, Steinmetz formula:

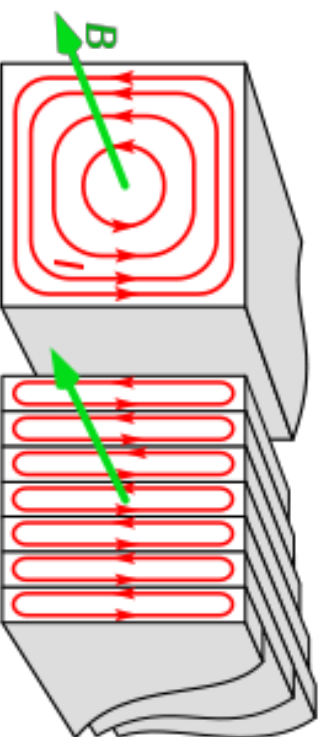
$$W_h = \eta B_{\max}^{1.6} f V \text{ (watts)}$$

where, η = Steinmetz hysteresis constant

V = volume of the core in m^3

Eddy current loss: When the armature core rotates in the magnetic field, an emf is also induced in the core (just like it induces in armature conductors), according to the [Faraday's law of electromagnetic induction](#). Though this induced emf is small, it causes a large current to flow in the body due to the low resistance of the core. This current is known as eddy current. The power loss due to this current is known as eddy current loss.

$$P_e = K_e * B_{max}^2 * f^2 * t^2 * V$$



To reduce eddy current loss in the core, the core is made up of varnished, laminated steel sheets instead of a solid core.

Mechanical and Brush Friction Losses

Friction loss occur at the bearing and brush friction loss occurs in wound rotor induction motor.

These losses are zero at start and with increase in speed these losses increases.

$$W_f = C_f N$$

Windage Loss occurs between a rotating solid part of the machine and the air around the surface of the rotor.

$$W_w = C_w N^2$$

The total mechanical loss = $W_m = C_f N + C_w N^2$

In three phase induction motor the speed usually remains constant. Hence these losses almost remains constant.

Electrical Losses

Stator copper loss = $3I_1^2 R_1$

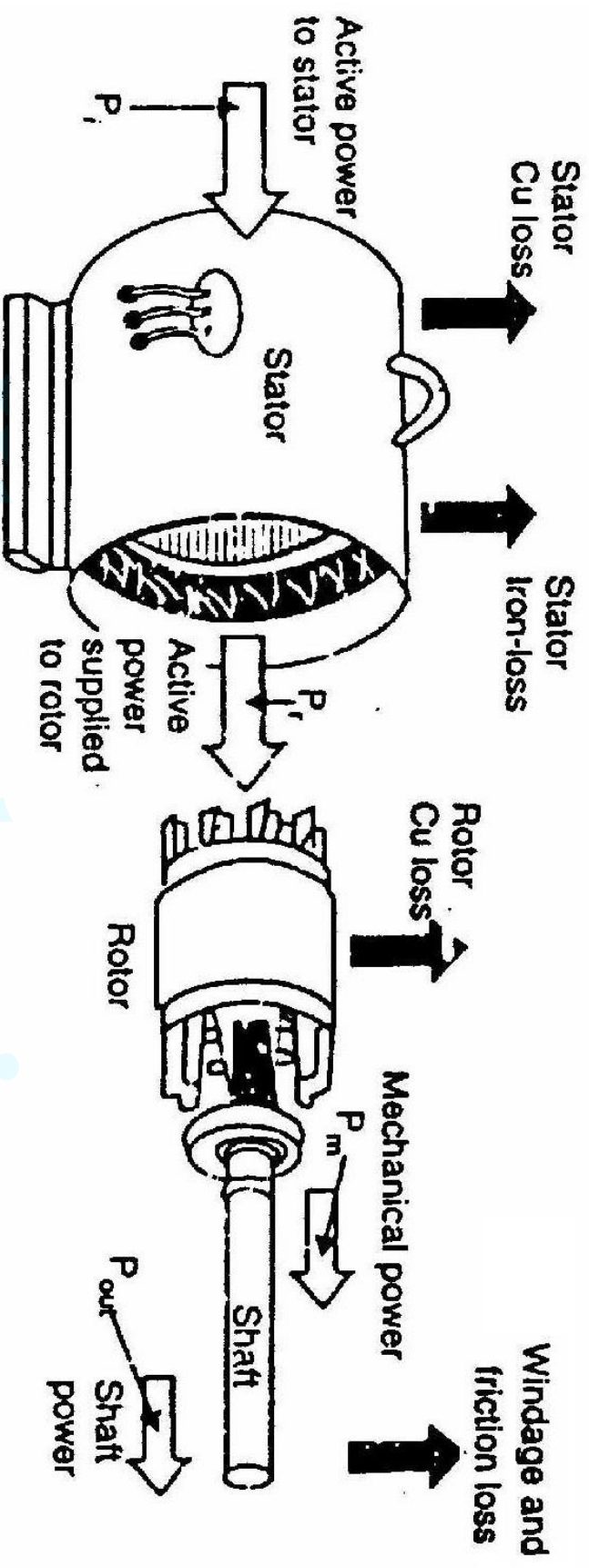
Rotor copper loss = $3I_2^2 R_2$

1. Fixed losses

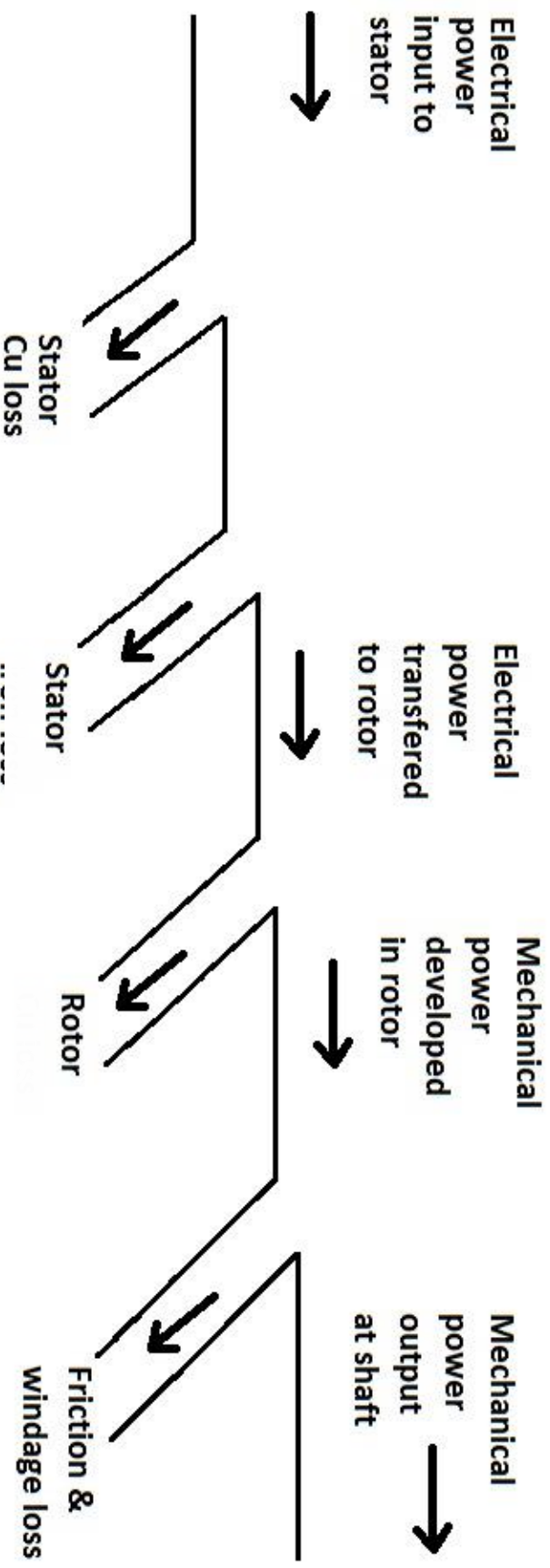
- a) Stator iron losses
- b) Friction & windage losses
- Rotor iron losses are negligible

2. Variable losses

- a) Stator Cu loss
- b) Rotor Cu loss



Power flow diagram



POWER STAGES IN AN INDUCTION MOTOR

Let, P_{in} = electrical power supplied to the stator of three phase induction motor,

V_L = line voltage supplied to the stator of three phase induction motor,

I_L = line current,

$\cos\phi$ = power factor of the three phase induction motor.

Electrical power input to the stator, $P_{in} = \sqrt{3}V_L I_L \cos\phi$

A part of this power input is used to supply **stator losses** which are **stator iron loss and stator copper loss**. The remaining power i.e (input electrical power – stator losses) are supplied to rotor as rotor input.

So, **rotor input** $P_2 = P_{in} - \text{stator losses}$ (stator copper loss and stator iron loss).

- Now, the rotor has to convert this rotor input into mechanical energy but this complete input cannot be converted into mechanical output as it has to supply rotor losses.
- The rotor losses are of two types **rotor iron loss and rotor copper loss**.
- Since the iron loss depends upon the rotor frequency, which is very small when the rotor rotates, so it is usually neglected.
- **So, the rotor has only rotor copper loss.**
- Therefore the rotor input has to supply these rotor copper losses.
- After supplying the rotor copper losses, the remaining part of Rotor input, P_2 is converted into mechanical power, P_m .

Let P_c be the rotor copper loss,

I_2 be the rotor current under running condition,

R_2 is the rotor resistance,

P_m is the gross mechanical power developed

$$P_c = 3I_2^2 R_2$$

$$P_m = P_2 - P_c$$

Now this mechanical power developed is given to the load by the shaft but there occur some mechanical losses like friction and windage losses.

So, the gross mechanical power developed has to be supplied to these losses. Therefore the net output power developed at the shaft, which is finally given to the load is P_{out} .

$P_{out} = P_m - \text{Mechanical losses}$ (friction and windage losses).

P_{out} is called the shaft power or useful power.

Three phase induction motor efficiency,

$$\eta = \frac{\text{power developed at shaft}}{\text{electrical input to the motor}}$$

$$\eta = \frac{P_{out}}{P_m}$$

Relation between Torque, Mechanical Power and Rotor Output

$$\text{Gross rotor output, } P_m = \frac{2\pi NT_g}{60}$$

$$\therefore \text{Gross Torque, } T_g = \frac{P_m}{2\pi N/60}$$

If there is no Cu losses in the rotor, then rotor output will equal rotor input and the rotor will run at synchronous speed.

$$\text{Rotor input, } P_2 = \frac{2\pi N_s T_g}{60}$$

$$\therefore \text{Rotor copper loss} = \text{Rotor input} - \text{Rotor output}$$

$$= \frac{2\pi T_g}{60} \times (N_s - N)$$

$$\frac{\text{Rotor copper loss}}{\text{Rotor input}} = \frac{N_s - N}{N_s} = s \quad \therefore \text{Rotor copper loss} = s \times \text{Rotor input} = s \times P_2$$

Gross rotor output, P_m = Rotor input – Rotor Cu loss

= Rotor input – $s \times$ Rotor input

= $(1 - s) \times$ Rotor input = $(1 - s) \times P_2$

$$\frac{\text{Gross rotor output}}{\text{Rotor input}} = 1 - s = \frac{N}{N_s}$$

$$\frac{\text{Rotor copper loss}}{\text{Gross rotor output}} = \frac{s}{1 - s}$$

$$\text{Rotor efficiency} = \frac{\text{Gross rotor output}}{\text{Rotor input}} = 1 - s$$

1. A 6 pole, 50 Hz, 3 phase induction motor runs at 960 r.p.m. when the torque on the shaft is 200 Nm. If the stator losses are 1500 W and mechanical losses in the shaft is 200 W. Calculate (i) rotor copper loss and (ii) efficiency of the motor.

Solution :

$$\text{Synchronous speed, } N_s = \frac{120 f}{P} = \frac{120 \times 50}{6} = 1000 \text{ r.p.m.}$$

$$\text{Slip, } s = \frac{N_s - N}{N_s} = \frac{1000 - 960}{1000} = 0.04 = 4\%$$

$$\begin{aligned} \text{Shaft power} &= \frac{2\pi N T_{sh}}{60} \\ &= \frac{2 \times 3.14 \times 960 \times 200}{60} \\ &= 20096 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Gross mechanical power, } P_m &= \text{shaft power} + \text{mechanical losses} \\ &= 20096 + 200 \\ &= 20296 \text{ W} \end{aligned}$$

(i)

$$\begin{aligned}\text{Rotor copper loss} &= \frac{s}{1-s} \times P_m \\ &= \frac{0.04}{1-0.04} \times 20296 \\ &= 845.87 \text{ W}\end{aligned}$$

(ii)

$$\text{Power input to rotor, } P_2 = \frac{P_m}{1-s} = \frac{20296}{1-0.04} = 21141.67$$

Motor input, $P_1 = P_2 + \text{stator losses}$

$$\begin{aligned}&= 21141.67 + 1500 \\ &= 22641.67 \text{ W}\end{aligned}$$

$$\text{Efficiency, } \eta = \frac{\text{Shaft power}}{\text{Motor input}} \times 100$$

$$\begin{aligned}&= \frac{20096}{22641.67} \times 100 \\ &= 88.76\%\end{aligned}$$

2. A 440-V, 3- ϕ , 50-Hz, 4-pole, Y-connected induction motor has a full-load speed of 1425 rpm. The rotor has an impedance of $(0.4 + j4)$ ohm and rotor/stator turn ratio of 0.8. Calculate (i) full-load torque (ii) rotor current and full-load rotor Cu loss (iii) power output if windage and friction losses amount to 500 W (iv) maximum torque and the speed at which it occurs (v) starting current and (vi) starting torque.

Solution.

$$N_s = 120 \times 50/4 = 1500 \text{ rpm} = 25 \text{ rps}, s = 75/1500 = 0.05$$

$$E_1 = 440/1.73 = 254 \text{ V/phase}$$

(i) Full-load

$$T_f = \frac{3}{2\pi \times 25} \times \frac{0.05 (0.8 \times 254)^2 \times 0.4}{(0.4)^2 + (0.05 \times 4)^2}$$
$$= \mathbf{78.87 \text{ N-m}}$$

(ii)

$$I_r = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{sKE_1}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{0.05 \times (0.8 \times 254)}{\sqrt{(0.4)^2 + (0.05 \times 4)^2}}$$
$$= \mathbf{22.73 \text{ A}}$$

Total Cu loss

$$= 3I_r^2 R = 3 \times 22.73^2 \times 0.4 = \mathbf{620 \text{ W}}$$

(iii) Now,

$$P_m = 2\pi NT = 2\pi \times (1425/60) \times 78.87 = 11,745 \text{ W}$$

∴

$$P_{\text{out}} = P_m - \text{windage and friction loss} = 11,745 - 500 = \mathbf{11,245 \text{ W}}$$

(iv)

For maximum torque, $s = R_2/X_2 = 0.4/4 = 0.1$

∴

$$T_{\text{max}} = \frac{3}{2\pi \times 25} \times \frac{0.1 \times (0.8 \times 254)^2 \times 0.4}{(0.4)^2 + (0.1 \times 4)^2} = \mathbf{98.5 \text{ N-m}}$$

Since

$$s = 0.1, \text{ slip speed} = s N_s = 0.1 \times 1500 = 150 \text{ rpm.}$$

$$\therefore \text{Speed for maximum torque} = 1500 - 150 = \mathbf{1350 \text{ rpm.}}$$

(v) starting current

$$= \frac{E_2}{\sqrt{R_2^2 + X_2^2}} = \frac{KE_1}{\sqrt{R_2^2 + X_2^2}} = \frac{0.8 \times 254}{\sqrt{0.4^2 + 4^2}} = \mathbf{50.5\ A}$$

(vi) At start,

$$s = 1, \text{ hence}$$

$$T_{st} = \frac{3}{2\pi \times 25} \times \frac{(0.8 \times 254)^2 \times 0.4}{(0.4)^2 + 4^2} = \mathbf{19.5\ N\cdot m}$$

It is seen that as compared to full-load torque, the starting torque is much less-almost 25 per cent.

3. The power input to a 500 V, 50-Hz, 6-pole, 3-phase induction motor running at 975 rpm is 40 kW. The stator losses are 1 kW and the friction and windage losses total 2 kW. Calculate : (i) the slip (ii) the rotor copper loss (iii) shaft power and (iv) the efficiency

Solution. (i) $N_s = 120 \times 50/6 = 1000$ rpm; $s = (100 - 975)/1000 = 0.025$ or 2.5%

(ii) Motor input $P_1 = 40$ kW; stator loss = 1 kW; rotor input $P_2 = 40 - 1 = 39$ kW

$$\therefore \text{rotor Cu loss} = s \times \text{rotor input} = 0.025 \times 39 = \mathbf{0.975 \text{ kW}}$$

(iii) $P_m = P_2 - \text{rotor Cu loss} = 39 - 0.975 = 38.025$ kW

$$P_{out} = P_m - \text{friction and windage loss} = 38.025 - 2 = \mathbf{36.025 \text{ kW}}$$

(iv) $\eta = P_{out}/P_1 = 36.025/40 = 0.9$ or **90%**

Equivalent circuit of an Induction motor

In an Induction motor, power is transferred from stator to rotor by induction.

i.e, Induction motor can be considered as a Transformer with rotating secondary winding. Also secondary winding is permanently short circuited.

In a transformer load is electrical, but in an induction motor load is mechanical (which is represented by an equivalent fictitious resistance)

So equivalent circuit of transformer can be represented by using a transformer equivalent circuit

Let K be the transformation ratio

The per phase equivalent circuit of induction motor w.r.to stator is shown in figure

V_1 = stator supply voltage/phase

R_1, X_1 = stator resistance & reactance/phase

R_2, X_2 = rotor resistance & reactance/phase

R_0, X_m = No load resistance & reactance/phase

R_L = per phase Electrical equivalent of mechanical power developed in rotor

I_1 = Stator current/phase

I_2 = rotor current/phase

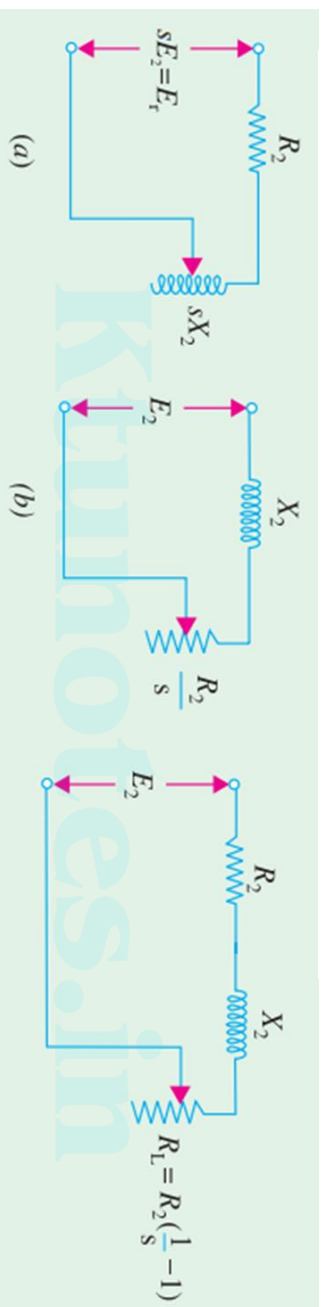
I_0 = No load current/phase



Basic Equivalent Circuit of I.M.

Consider the rotor side,
When motor is loaded, the rotor current I_2 is given by

$$I_2 = \frac{s \frac{E_2}{\sqrt{R_2^2 + (sX_2)^2}}}{\sqrt{[(R_2/s)^2 + X_2^2]}} = \frac{E_2}{\sqrt{[(R_2/s)^2 + X_2^2]}}$$

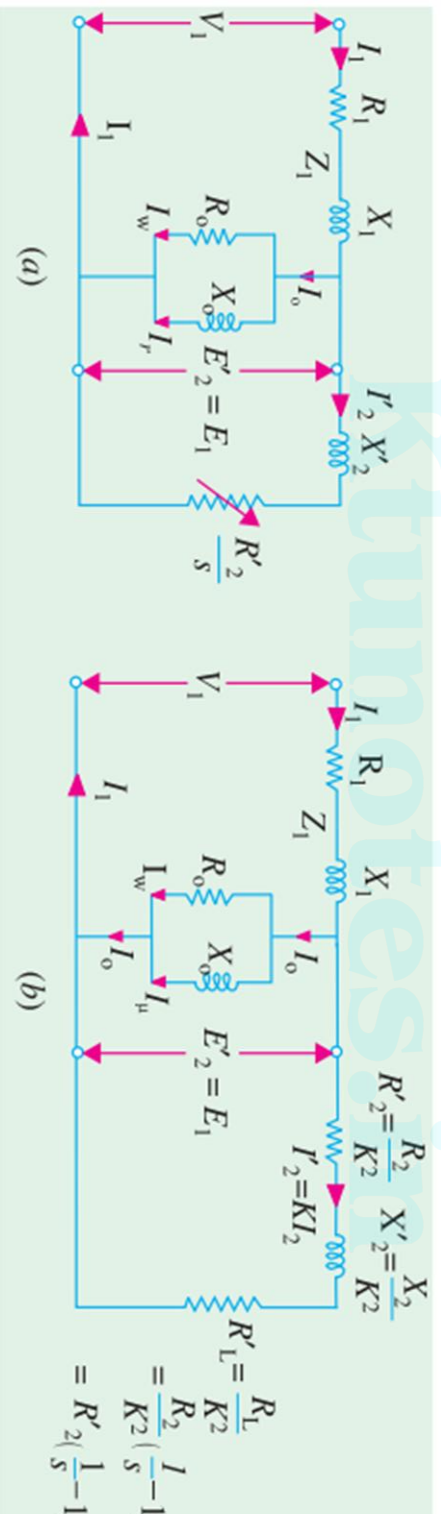


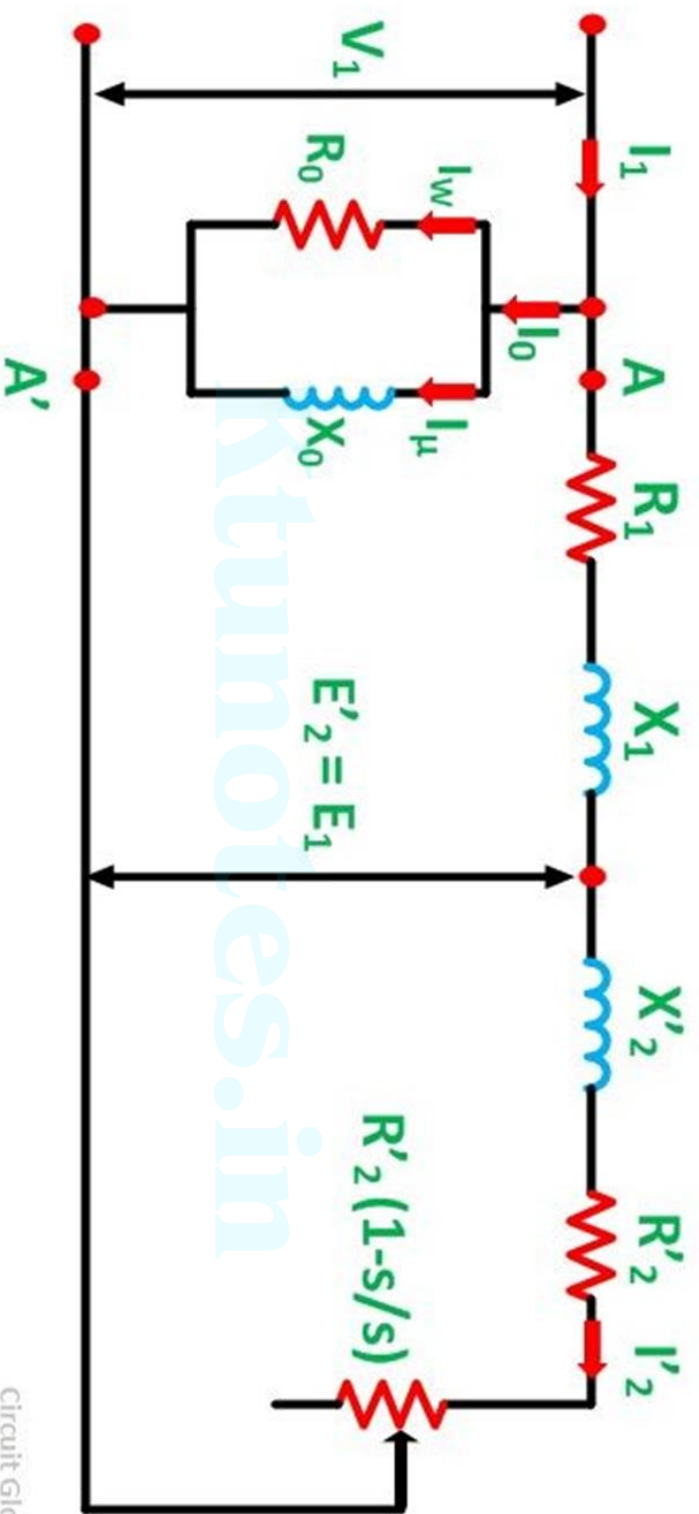
(i) the first part R_2 is the rotor resistance itself and represents the rotor Cu loss.

(ii) the second part is $R_2\left(\frac{1}{s} - 1\right)$

This is known as the load resistance R_L and is the electrical equivalent of the mechanical load on the motor. In other words, the mechanical load on an induction motor can be represented by a non-inductive resistance of the value $R_2\left(\frac{1}{s} - 1\right)$

As in the case of a transformer in this case also, the secondary values may be transferred to the primary and vice versa. when shifting impedance or resistance from secondary to primary, it should be **divided by K^2** whereas current should be **multiplied by K** . The equivalent circuit of an induction motor where all values have been referred to primary i.e. stator is shown in Fig.





Circuit Globe

From equivalent circuit

$$\text{Rotor cu loss} = P_{cu} = 3I_2'^2 R_2'$$

$$\text{Rotor output} = 3I_2'^2 R_L' = P_{cu} = 3I_2'^2 R_2' \left(\frac{1}{s} - 1 \right) =$$

$$\text{Mechanical power developed} (P_m)$$

$$\text{Rotor input} = \text{Rotor Cu Loss} + \text{Rotor output} = P_{cu} + P_m = \frac{3I_2'^2 R_2'}{s}$$

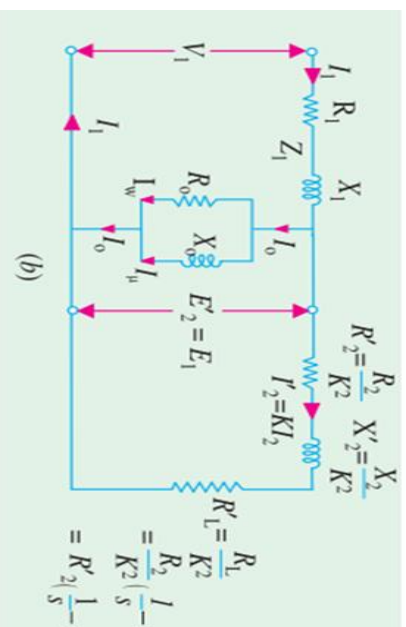
$$\text{Rotor input} = P_2$$

$$\text{Rotor cu loss} = sP_2$$

$$\text{Mechanical power developed} = P_m = (1 - s)P_2$$

$$\text{Shaft output} = P_{out} = P_m - \text{Mechanical losses}$$

Power Balance Equations



Input power = $3 V_1 I_1 \cos \phi_1$; stator core loss = $I_w^2 R_o$; stator Cu loss = $3 I_1^2 R_1$

Power transferred to rotor = $3 I_2'^2 R_2'/s$; Rotor Cu loss = $3 I_2'^2 R_2'$

Mechanical power developed by rotor (P_m) or gross power developed by rotor (P_g)

$$\begin{aligned}
 &= \text{rotor input} - \text{rotor Cu losses} \\
 &= 3 I_2'^2 R_2' / s - 3 I_2'^2 R_2' = 3 I_2'^2 R_2' \left(\frac{1-s}{s} \right) \text{ Watt}
 \end{aligned}$$

If T_g is the gross torque* developed by the rotor, then

$$T_g \times \omega = T_g \times 2\pi \frac{N}{60} = 3 I_2'^2 R_2' \left(\frac{1-s}{s} \right)$$

$$\therefore T_g = \frac{3 I_2'^2 R_2' \left(\frac{1-s}{s} \right)}{2\pi N / 60} \text{ N-m}$$

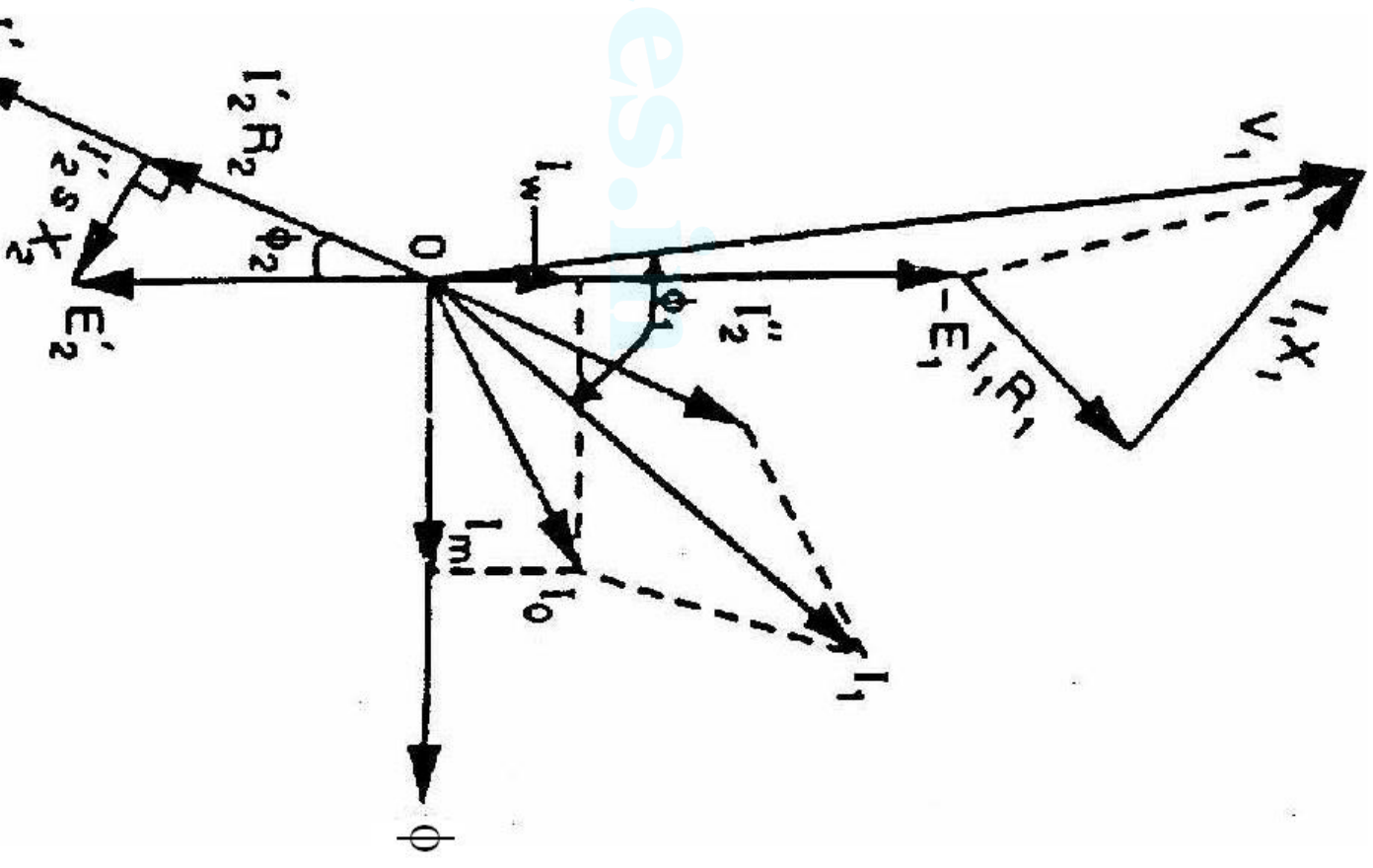
Now, $N = N_s(1-s)$. Hence gross torque becomes

$$T_g = \frac{3 I_2'^2 R_2' / s}{2\pi N / 60} \text{ N-m} = 9.55 \times \frac{3 I_2'^2 R_2' / s}{N_s} \text{ N-m}$$

Phasor diagram of Induction motor

motor

Phasor diagram can be drawn based on the equivalent circuit of Induction motor



Tests for determination of equivalent circuit of Induction Motor

1. Stator Resistance Test

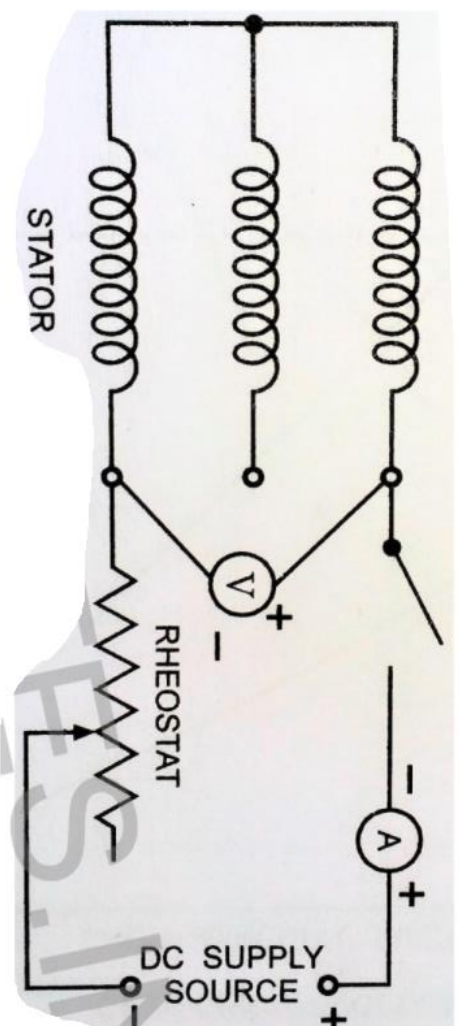
2. No-load Test

Stator core loss, friction & windage losses can be determined.

3. Blocked Rotor Test

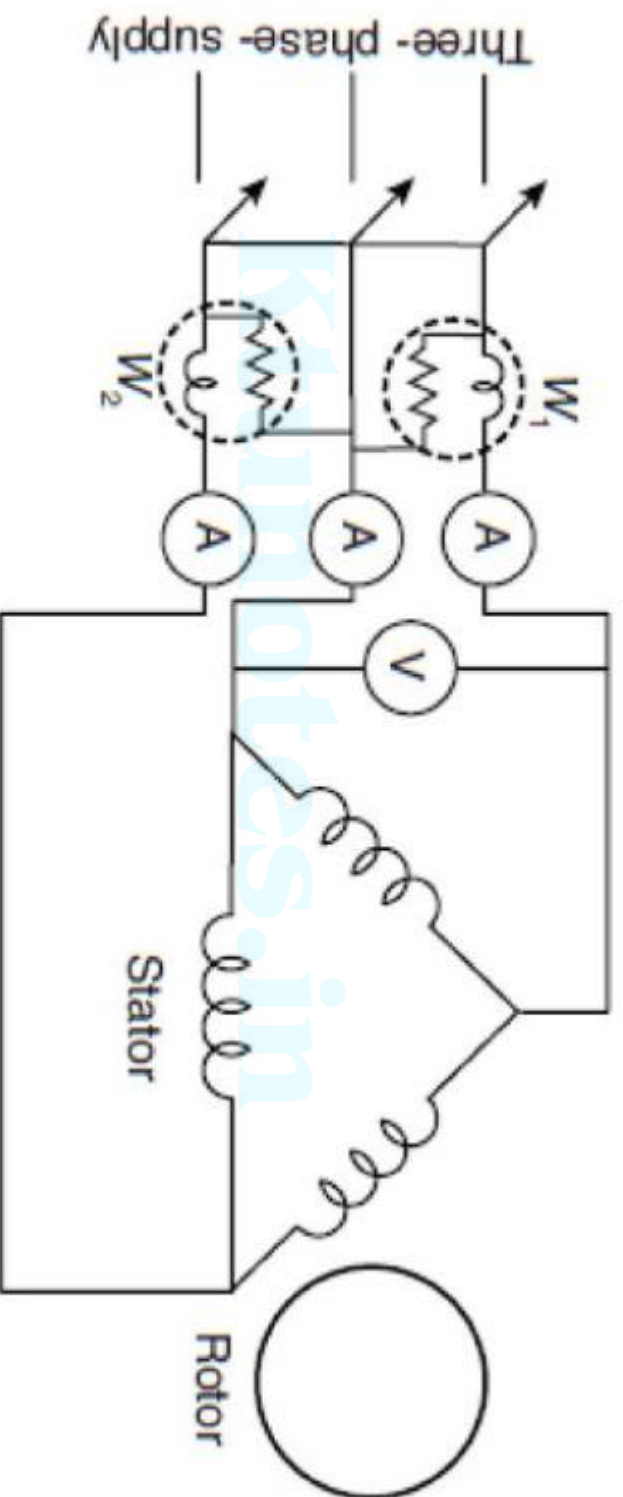
Stator & rotor copper losses can be determined.

1. Stator Resistance Test



- R₁ can be measured by Voltmeter-Ammeter Method.
- DC resistance between a pair of terminals is measured.
- Resistance of each phase, $R_{1dc} = \frac{V}{2I}$ ohms
- Effective AC Resistance per phase(R_1) = (1.25 to 1.75)times R_{1dc}

No-load Test



2. No load test (Equivalent to No load test in transformer)

- During this test, Induction motor is made to run at no load by applying rated voltage & frequency supply
- The connection diagram is shown in figure

Voltmeter reading = V_{OL} = No load line voltage

Ammeter reading = I_{OL} = No load line current

Wattmeter reading, $W_1 + W_2 = 3$ phase power input at no load = P_o

No load voltage/phase = $V_{OPH} = V_{OL}/(\sqrt{3})$

No load current/phase = $I_{OPH} = I_{OL}$

No load power input = $P_o = 3V_{OPH}I_{OPH}\cos\phi_o$

$$\cos\phi_0 = \text{No load p.f} = \frac{P_0}{3V_{OPH}I_{OPH}}$$

$$I_{MPH} = I_{OPH} \sin\phi_0$$

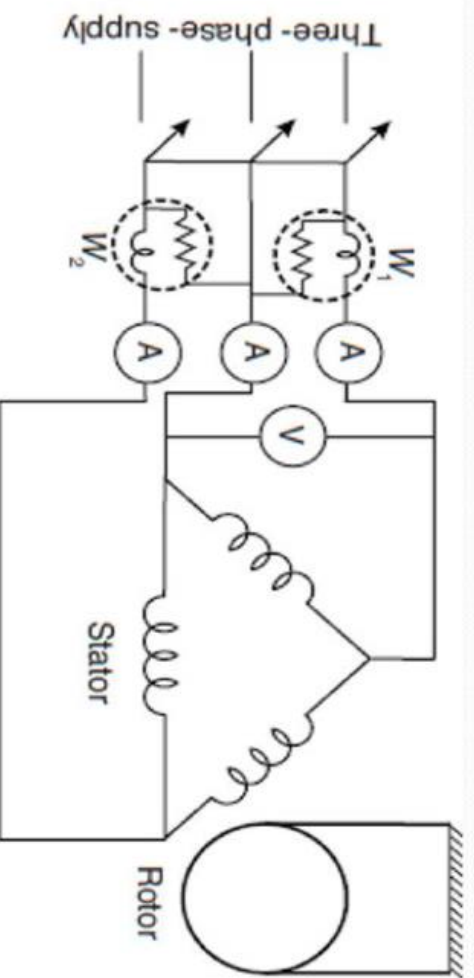
$$I_{WPH} = I_{OPH} \cos\phi_0$$

$$R_0 = V_{OPH}/I_{WPH}$$

$$X_0 = V_{OPH}/I_{MPH}$$

3. Blocked rotor test (Equivalent to Short circuit test in transformer)

- During this test, rotor is blocked from rotation by using hand or by using a brake drum & belt arrangement
- Now the stator is supplied from a variable voltage supply
- The stator applied voltage is increased in steps till the stator current reaches rated value
- Now the meter readings are taken
- The connection diagram is shown in figure



Voltmeter reading = V_{scl} = Blocked rotor/Short circuit line voltage

Ammeter reading = I_{scl} = Short circuit line current

Wattmeter reading, $W_1 + W_2 = 3$ phase power input at short circuit = P_{sc}

Short circuit voltage/phase = $V_{scph} = V_{scl}/(\sqrt{3})$

Short circuit current/phase = $I_{scph} = I_{scl}$

Short circuit power input = $P_{sc} = 3V_{scph}I_{scph}\cos\phi_{sc} = 3I_{scph}^2R_{01}$

$$\cos\phi_{sc} = \text{Short Circuit p.f} = \frac{P_{sc}}{3V_{scph}I_{scph}} \quad R_{01} = \frac{P_{sc}}{3I_{scph}^2}$$

$$Z_{01} = V_{scph}/I_{scph}, \quad X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$$

$X_{01} = X_1 + X_2'$. Normally stator & rotor reactance/phase is assumed to be constant. i.e, $X_1 = X_2 = X_{01}/2$

$R_{01} = R_1 + R_2'$. Stator resistance/phase is measured by Ammeter-Voltmeter method.

Rotor resistance/phase, $R_2' = R_{01} - R_1$

